



REPUBLIC OF TUNISIA
Ministry of Higher Education and Scientific Research
University of Carthage
National Institute of Applied Sciences and Technology



Graduation Project

to obtain the

National Engineering Diploma

in Applied Sciences and Technology

Specialty: Instrumentation and Industrial Maintenance

Design and Implementation of a Comprehensive OEE Framework for an Automated Transport System

Prepared by

Oumayma LAADHAR

Hosted by TII KAMAG



Defended on XX/XX/2026 in front of a jury composed of:

XX President of the jury

XX Reviewer

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Mr. Martin TEUCHLER Supervisor at the Company

School year: 2025/2026



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Dedication

To my parents,

who have supported me from the very beginning, stood by my side through every decision, and always believed in me. Your love, sacrifices, and encouragement have been the foundation of everything I have achieved.

To my siblings,

who have always been there for me, bringing support, strength, and companionship throughout this journey.

To my friends,

who shared this journey with me, supported me through the difficult moments, celebrated the good ones, and made this chapter of my life even more meaningful.

This achievement is as much yours as it is mine.

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List of Acronyms

ADS Automated Driving System. 15, 17

AGV Automated Guided Vehicle. 1, 13, 14

AMR Autonomous Mobile Robot. 13, 14

AV Autonomous Vehicle. 1, 6, 7, 9

DUR Dispatch Utilisation Rate. 28, 29, 55, 64

ESR Execution Speed Rate. 28, 29, 55, 64, 74, 75

GEE Global Equipment Effectiveness. 30, 31, 58

JIPM Japan Institute of Plant Maintenance. 10

KAMAG PM KAMAG PrecisionMover. 3–5

KAMAG PT Kamag Precision Tractor. 3

KPI Key Performance Indicator. 7, 15, 16, 20, 22

LSL Lower Specification Limit. 19

MES Manufacturing Execution System. 12

MTBF Mean Time Between Failures. 18, 20

MTTR Mean Time To Repair. 18, 20

ODD Operational Design Domain. 1, 4, 15, 17, 18, 20, 22, 23, 46, 48, 50, 51

OOE Overall Equipment Effectiveness. v, 1, 4, 7, 9, 10, 14–16, 18, 22, 23, 30, 31, 44, 46–80

RAM Reliability, Availability, and Maintainability. 12

RMF Road Map File. 57, 58, 70

SOTIF Safety of the Intended Functionality. 17

SPC Statistical Process Control. 19

TEEP Total Effective Equipment Performance. 30, 31, 58

TPM Total Productive Maintenance. 10

USL Upper Specification Limit. 19

Introduction

The increasing automation of logistics operations is transforming the way industrial transport systems are designed and operated. Autonomous Vehicles (AVs) are becoming an integral part of modern logistics yards, where they perform repetitive transport tasks with the objective of improving efficiency, safety, and operational continuity. Unlike conventional Automated Guided Vehicles (AGVs) operating in highly controlled environments, autonomous yard vehicles operate in dynamic, semi-structured environments shared with human workers, manually driven vehicles, and subject to changing environmental conditions. These characteristics introduce new challenges for monitoring and evaluating operational performance.

OEE is a widely adopted methodology for assessing industrial performance. By combining the three dimensions of Availability, Performance, and Quality, OEE provides a concise measure of how effectively equipment converts operating time into productive work. However, its original formulation was developed for stationary manufacturing equipment and does not adequately reflect the characteristics of autonomous logistics operations, where mission variability, remote supervision, and Operational Design Domain (ODD) constraints significantly influence system behaviour.

This work was carried out at TII KAMAG within an industrial project focused on autonomous yard logistics. Its objective is to design and implement an OEE-based performance measurement framework adapted to autonomous vehicle operations. The proposed framework preserves the fundamental principles of classical OEE while extending them to account for the specific characteristics of autonomous yard logistics through an adapted time hierarchy, autonomous-vehicle-specific loss classification, ODD-aware time accounting, and complementary engineering indicators supporting the evaluation of reliability, operational capacity, supervision requirements, execution quality, and environmental influences.

This work is organised as follows. The first chapter presents the industrial context, the autonomous transport system, the available operational data, and the relevant state of the art. The second chapter introduces the proposed OEE framework, including its measurement principles, time hierarchy, indicators, and canonical data model. The third chapter describes its implementation and evaluates the framework using industrial data and controlled synthetic datasets, including a mission-phase diagnostic extension. This work concludes by summarising the main findings, limitations, and perspectives for future deployment.

Chapter 1

Industrial Context and State of the Art

1.1 Introduction

Autonomous yard-logistics systems integrate industrial transport, automated driving, remote supervision, and changing environmental conditions. Evaluating their performance requires considering more than the vehicle alone. It must also account for the logistics process, operational constraints, and available data. This chapter therefore examines the industrial context and relevant state of the art.

It first presents the host company and project environment. It then describes the autonomous yard-logistics system, its operating area, system actors, transfer workflow, mission phases, architecture, and data sources.

The theoretical foundations of Overall Equipment Effectiveness are then reviewed. The discussion focuses on its classical model, applications beyond manufacturing, and limitations for autonomous systems. The chapter also examines autonomous-vehicle metrics, ODD, SOTIF, reliability indicators, and process capability. Finally, the chapter identifies the research gap and presents the project methodology.

1.2 Industrial and Project Context

1.2.1 Presentation of TII KAMAG

This graduation project was carried out at TII KAMAG, a subsidiary of the TII Group (Transporter Industry International Group), a globally recognized manufacturer of heavy-duty transport vehicles and specialized logistics solutions. Founded in 1969, TII KAMAG has evolved from a small engineering company into a leading provider of innovative transport systems designed to meet the demanding requirements of heavy industries.[1] [2]



Figure 1.1: TII KAMAG Logo



Figure 1.2: TII Group Logo

TII KAMAG operates in several industrial domains, including automotive, aerospace, and logistics solutions for heavy-goods transport. As a transport-technology provider, the company is developing autonomous driving systems to improve logistics processes in heavy industry. This work follows two complementary directions: improving existing vehicle systems and developing solutions that can perform complex tasks autonomously in demanding environments.

TII KAMAG offers a range of industrial transport solutions, focusing on advanced logistics and material-handling applications. One key product, the Kamag Precision Tractor (KAMAG PT), is a specialized terminal vehicle for transport and handling operations in logistics environments. The KAMAG PrecisionMover (KAMAG PM) complements this solution by handling and transporting swap bodies and semi-trailers within logistics hubs and industrial yards. Logistics companies such as DACHSER have used this application for many years, demonstrating its industrial relevance. [2]

These vehicles reflect TII KAMAG's focus on intelligent technologies and autonomous driving to improve efficiency and reliability in demanding environments. The company also provides fleet management, operator training, and maintenance services to support reliable operation for its international customers.



Figure 1.3: Kamag Precision Tractor[2]



Figure 1.4: Kamag Precision Mover[2]

TII KAMAG operates from an industrial site comprising multiple production facilities and technical departments dedicated to the development of transport technologies. The company employs approximately 350 people who contribute to the design, manufacturing, and continuous improvement of specialized transport solutions for customers worldwide.



Figure 1.5: TII KAMAG Campus[2]

1.2.2 Project Context and Objectives

The development of autonomous transport solutions creates a need for suitable methods to evaluate their operational effectiveness. Unlike stationary production equipment, autonomous yard vehicles execute variable transport missions under changing traffic and environmental conditions. These characteristics make the direct application of classical OEE insufficient for identifying and interpreting the main sources of operational loss.

Within this industrial context, the objective of this graduation project is to **design and implement an OEE-based performance measurement framework adapted to autonomous yard logistics**. The work aims to define suitable Availability, Performance, and Quality metrics, account for ODD-related constraints. It also provides complementary indicators for deeper operational diagnosis. The framework is supported by a structured data model and implemented using the available operational data.

1.3 Autonomous Yard-Logistics System

The system under study operates within the DACHSER yard, where autonomous vehicles perform swap body transfer operations in a defined industrial logistics area.

1.3.1 Operating Environment and System Actors

The yard is a semi-structured environment with predefined driving paths connecting multiple parking locations. Parking locations may contain swap bodies, standardized load units supported by folding legs, or no load unit. The area is shared with pedestrians and manually operated vehicles, so the autonomous system must navigate intersections, crossing areas, and changing operational conditions.

Three actors are involved in the operation: the autonomous vehicle (KAMAG PM), which executes transport missions; the remote supervisor or safety operator, who monitors operation,

manages protective stops, validates specific manoeuvres, and intervenes when necessary; and the yard operations team, which manages transport orders, mission coordination, and operational exceptions.

1.3.2 Swap Body Transfer Workflow

The transfer is the basic unit of operation and represents the relocation of one swap body from a source location to a destination location. It may involve two missions: a pick-up mission, during which the KAMAG PM travels to the source location, positions itself beneath the swap body, and lifts it; and a set-down mission, during which the loaded vehicle transports the swap body to the destination and unloads it.

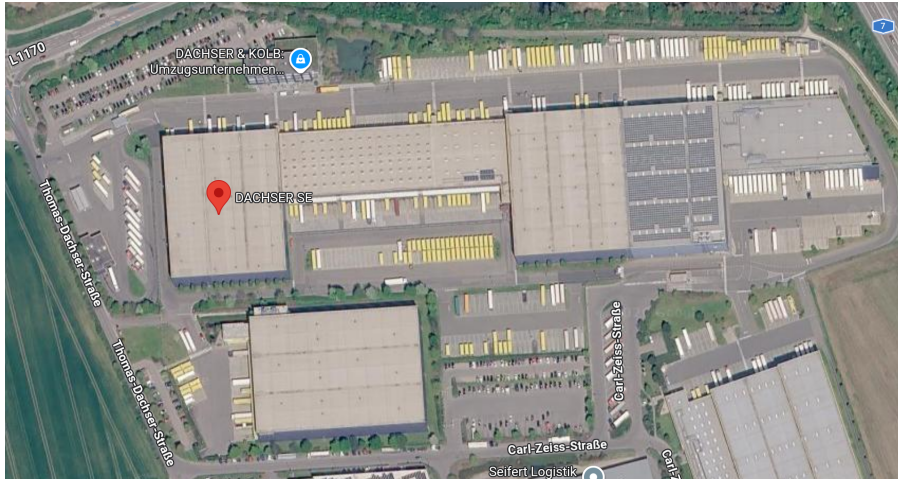


Figure 1.6: The Dachser Yard [3]

1.3.3 Operational Hierarchy

The logistics operation is structured into three levels: transfer, mission, and mission phase. A transfer represents the complete transport order from a source location to a destination. It is composed of one or two missions depending on the transfer scenario. Each mission describes a specific part of the transport operation, such as pick-up, set-down, or repositioning. Missions are further decomposed into elementary mission phases that describe the individual actions executed by the vehicle.

1.3.4 Mission Phases and Transfer Scenarios

The vehicle operation is divided into eight mission phases, denoted MP1 to MP8. These phases cover path following, manoeuvring, insertion, load handling, extraction, and return to the predefined path. Table 1.1 summarises their definitions, while Figure 1.7 illustrates the phase sequence.

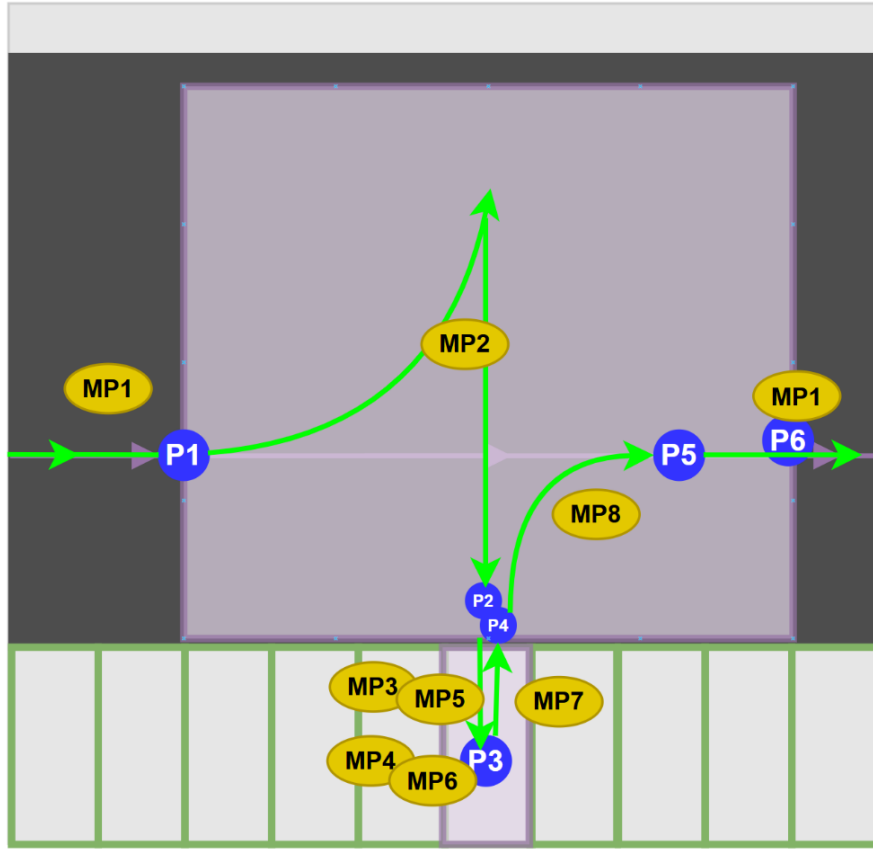


Figure 1.7: Mission Phases Representation

Table 1.1: Canonical Mission Phases

Phase	Description	Navigation mode	Detailed description
MP1	Follow the predefined path	follower	The AV drives forward or backward along the predefined path to P1
MP2	Manoeuvre toward pre-insertion point	manoeuvre	From P1 to P2
MP3	Insert into spot under swap body	manoeuvre	From P2 to P3
MP4	Pick up swap body	manoeuvre	At P3
MP5	Insert into empty spot	manoeuvre	From P2 to P3
MP6	Set down swap body	manoeuvre	At P3
MP7	Extract from spot	manoeuvre	From P3 to P4
MP8	Catch up the predefined path	manoeuvre	From P4 to P5

A transfer is classified by whether each of its two missions requires long-distance travel or takes place in close proximity to the target spot:

Table 1.2: Transfer scenarios

Scenario	Mission-phase sequence
classic	Mission 1: PICK_UP MP7 → MP8 → MP1 → MP2 → MP3 → MP4 Mission 2: SET_DOWN MP7 → MP8 → MP1 → MP2 → MP5 → MP6
Close proximity pick up	Mission 1: CLOSE_PROXIMITY_PICK_UP MP7 → MP2 → MP3 → MP4 Mission 2: SET_DOWN MP7 → MP8 → MP1 → MP2 → MP5 → MP6
Close proximity set down	Mission 1: PICK_UP MP7 → MP8 → MP1 → MP2 → MP3 → MP4 Mission 2: CLOSE_PROXIMITY_SET_DOWN MP7 → MP2 → MP5 → MP6
Double close proximity	Mission 1: CLOSE_PROXIMITY_PICK_UP MP7 → MP2 → MP3 → MP4 Mission 2: CLOSE_PROXIMITY_SET_DOWN MP7 → MP2 → MP5 → MP6
park (GO_TO)	Mission 1: GO_TO MP7 → MP8 → MP1 → MP2 → MP5 <i>Single mission scenario</i>

1.3.5 System and Data Architecture

The autonomous yard logistics system consists of two layers: an onboard vehicle layer and an off-board data-management layer. During transport missions, the AV continuously generates data on vehicle states, mission execution, events, and outcomes.

At the vehicle level, onboard logging records each mission’s operational trace, including timestamps, mission phases, system events, and execution results. These records support the reconstruction of transport operations and the identification of performance losses.

The telemetry data is transmitted to off-board systems, where fleet information is consolidated into an Automatic Telemetry Dataset. This dataset provides the main input for computing Key Performance Indicators (KPIs) and evaluating OEE.

This architecture links autonomous operation to performance analysis: vehicle execution generates operational data, telemetry systems collect and organize it, and the resulting dataset supports the evaluation of autonomous yard effectiveness.

Autonomous Vehicle Data Collection Architecture

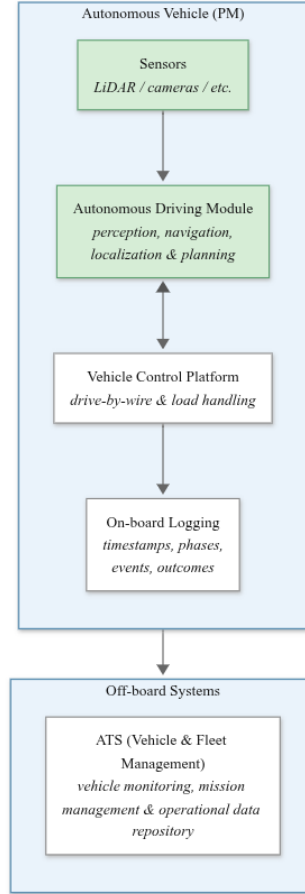


Figure 1.8: System Architecture

1.3.6 Available Data and Evaluation Constraints

This section describes the data foundation used to design and evaluate the proposed performance measurement framework.

- **Data sources:** The framework relies on two complementary data sources. The first is the automatic dataset, generated from vehicle operation and containing machine-recorded information such as timestamps, mission phases, operational events, and execution outcomes. The second is a manual field log maintained by operators, providing additional contextual information regarding interventions, exceptions, and operational events that cannot always be fully captured through automatic telemetry alone. These two sources are combined at the transfer level to create a unified structured dataset representing each transport operation. Real operational data from both sources was analysed during the design phase to identify the data fields and input structure required by the proposed framework, and to validate that the necessary information is practically collectable from the system.
- **Historical and synthetic data:** Due to limitations in data coverage and completeness, real records alone were insufficient for a comprehensive evaluation. A synthetic dataset

was therefore generated following the same input schema derived from the real data, enabling controlled and reproducible testing of the framework under predefined performance scenarios while remaining consistent with real operational characteristics.

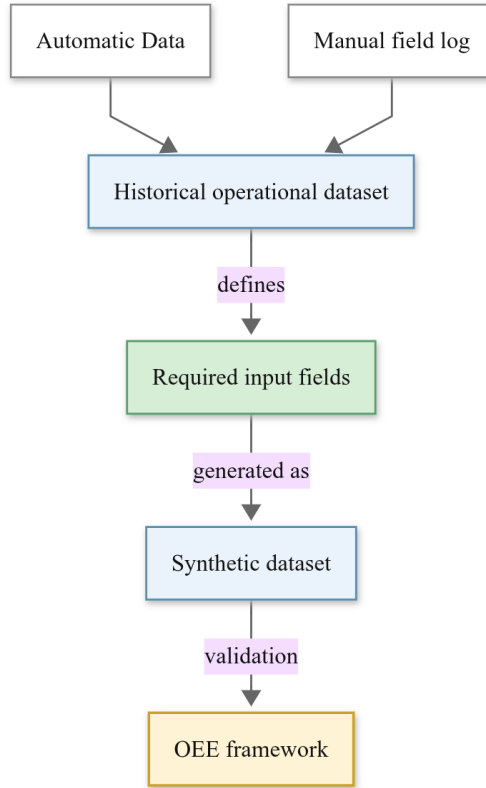


Figure 1.9: Data Provenance Flow

1.4 Industrial Performance-Measurement Problem

OEE was developed to evaluate stationary manufacturing equipment operating under repetitive and predictable conditions. Its classical formulation assumes fixed production cycles, clearly defined operating states, and losses categorized as availability, performance, and quality. Under these conditions, OEE effectively measures equipment utilization and production efficiency.

These assumptions do not fully apply to AVs operating in dynamic yard environments. Unlike industrial machines, AV missions are affected by travel distances, route configurations, traffic, loading and unloading operations, and interactions with people and infrastructure. Operational interruptions also cannot always be interpreted as equipment failures. An AV may stop because of safety constraints, remote-supervision requests, autonomy limitations, congestion, or environmental conditions while remaining technically functional.

Applying conventional OEE to autonomous yard operations may therefore obscure the true causes of performance losses by grouping fundamentally different events into identical categories. Mechanical failures, autonomy-related limitations, safety interventions, and environmental constraints require separate interpretation to support effective operational decisions.

Current autonomous-driving metrics, such as disengagement rate and intervention frequency, primarily evaluate autonomy and safety. However, they do not measure the overall effectiveness of logistics operations. Traditional OEE, measures equipment efficiency but lacks the operational states and loss categories required for autonomous vehicles. This creates a measurement gap: existing approaches cannot precisely determine the useful transport work performed by an autonomous fleet.

A dedicated performance-measurement framework is therefore required for autonomous yard logistics. Such a framework should preserve the industrial relevance of OEE while adapting its time hierarchy, performance indicators, and loss classification to mission variability, autonomy-specific events, supervision requirements, and operational constraints. This work addresses this need by developing an OEE-based performance-measurement approach for the autonomous transport system deployed in the DACHSER yard.

1.5 Overall Equipment Effectiveness Foundations

This section presents the foundations of OEE and its relevance to autonomous yard logistics. It reviews the classical OEE framework, its extension beyond manufacturing, and the assumptions underlying its main factors. It also introduces the Six Big Losses and relevant industrial standards. These foundations provide the basis for identifying the limitations that motivate the adapted framework proposed in this work.

1.5.1 Origins and Rationale of OEE

OEE is a key manufacturing metric used to quantify production efficiency by identifying and reducing operational losses. Seiichi Nakajima introduced the concept in 1988 as part of Total Productive Maintenance (Total Productive Maintenance (TPM)), a methodology developed by the Japan Institute of Plant Maintenance (Japan Institute of Plant Maintenance (JIPM)).[4] The objective of TPM is to maximize equipment effectiveness throughout its lifecycle through a comprehensive maintenance system involving all departments and employees.

Nakajima’s work provided a structured method for evaluating equipment performance beyond simple uptime. OEE combines equipment availability, operating speed, and product quality to reveal the capacity lost through inefficiencies. This lost capacity is often described as the “Hidden Factory.” By separating overall effectiveness into distinct factors, OEE supports targeted improvement actions and waste reduction.

1.5.2 Availability, Performance, and Quality

OEE is calculated as the product of three factors:

$$\text{OEE} = A \cdot P \cdot Q$$

Availability, Performance, and Quality represent different types of operational loss:

- **Availability** measures the proportion of scheduled operating time during which the equipment is running. It accounts for downtime losses such as breakdowns, setups, and adjustments.
- **Performance** compares the actual operating speed with the ideal cycle time. It accounts for losses caused by minor stops and reduced speed.
- **Quality** measures the proportion of good products in the total output. It accounts for losses caused by defects and rework.

1.5.3 Fixed-Cycle Assumption

Classical OEE relies on a fixed-cycle assumption. For a given product, the ideal production cycle is assumed to remain constant. This cycle provides the reference for measuring actual operating speed and calculating the Performance factor. Deviations from the ideal cycle, including minor stops and reduced speed, are therefore classified as performance losses.

This assumption is effective in stable and repetitive manufacturing environments. However, it becomes difficult to apply to dynamic systems whose mission durations and operating conditions vary. This limitation is particularly important when extending OEE to autonomous vehicles.

1.5.4 The Six Big Losses

Nakajima's OEE framework classifies production inefficiencies into the "Six Big Losses." [4] This taxonomy provides a structured method for identifying the main causes of equipment underutilization:

1. **Equipment Failure (Availability Loss):** unplanned downtime caused by breakdowns or technical faults.
2. **Setup and Adjustment (Availability Loss):** planned downtime required to prepare the machine for a different production run.
3. **Idling and Minor Stops (Performance Loss):** brief interruptions or jams that temporarily stop production.
4. **Reduced Speed (Performance Loss):** operation at a speed below the theoretical maximum.
5. **Process Defects (Quality Loss):** scrap or defective parts produced during steady-state operation.
6. **Reduced Yield (Quality Loss):** defective parts produced during startup or production transitions.

The Six Big Losses are highly effective for static and repetitive manufacturing processes because they provide a clear structure for continuous improvement. However, their direct application to autonomous yard logistics requires adaptation to mobile operation, variable missions, and autonomy-related events.

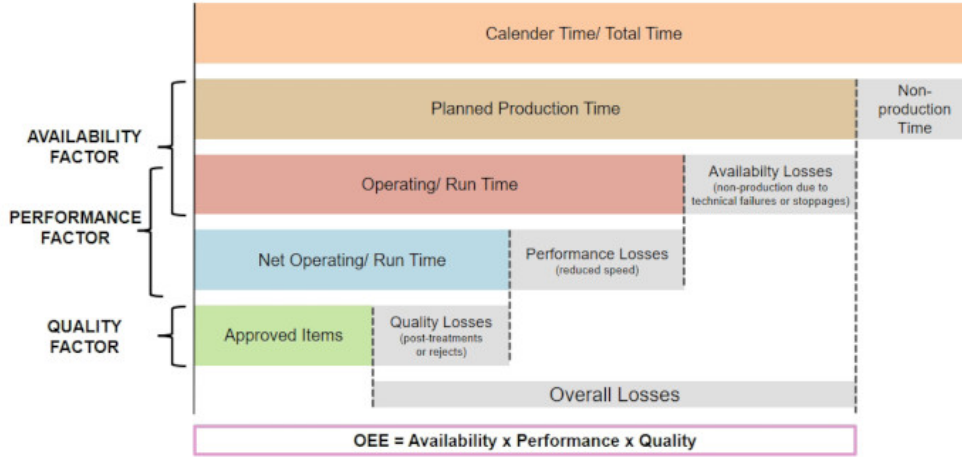


Figure 1.10: Major Losses and Computation of OEE[5]

1.5.5 Industrial Standards and Guidelines

Several standards and industrial guidelines support the consistent definition and measurement of OEE-related indicators:

- **ISO 22400-2:2014:** This standard defines KPIs for manufacturing operations management. It provides formal definitions for indicators such as Availability, which is calculated as the ratio of Operating Time to Planned Production Time.[6]
- **SEMI E10:** Developed for the semiconductor industry, this standard defines equipment states and provides a framework for measuring reliability, availability, and maintainability (Reliability, Availability, and Maintainability (RAM)). It distinguishes states such as Productive, Standby, and Unscheduled Downtime.[7] SEMI E79 extends this framework to define OEE for semiconductor manufacturing equipment.
- **VDMA guidelines:** Guidelines such as VDMA 66412 support the standardization of manufacturing execution system (Manufacturing Execution System (MES)) KPIs. They are generally aligned with ISO 22400 and provide guidance for data collection and analysis, with particular emphasis on technical availability.

1.6 OEE Beyond Manufacturing

While OEE originated in discrete manufacturing, its underlying principles of identifying, classifying, and quantifying operational losses have gradually been extended to other industrial

domains. As automation has expanded beyond traditional production lines, researchers have explored the application of OEE to logistics, warehousing, and mobile robotic systems. Although the fundamental objective remains the maximization of asset effectiveness, the transition from stationary production equipment to mobile autonomous systems requires substantial adaptation of both performance indicators and loss definitions.

1.6.1 OEE in Logistics and Material-Handling Systems

The application of OEE has progressively expanded from manufacturing equipment to automated material handling systems, including conveyor systems, sorting equipment, AGVs, and Autonomous Mobile Robots (AMRs). These technologies play an increasingly important role in modern intralogistics by automating material transportation and improving operational efficiency.

1.6.2 Adaptation of OEE Components to Mobile Systems

Existing research shows that the three classical OEE dimensions remain applicable to mobile robotic systems, but require reinterpretation. Availability reflects vehicle readiness and downtime, Performance focuses on mission execution efficiency and speed, and Quality considers successful and error-free mission completion. Thus, OEE shifts from measuring machine productivity to evaluating the effectiveness and utilization of autonomous material handling systems [8].

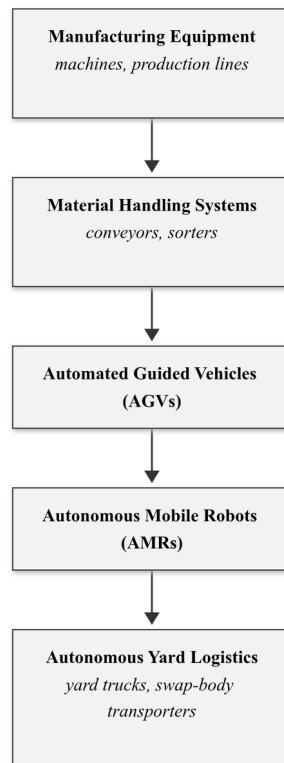


Figure 1.11: Evolution of OEE applications in logistics and mobile robotic systems.

1.6.3 Adaptation of the Six Big Losses

Applying OEE to mobile logistics also requires reinterpreting Nakajima’s Six Big Losses. The original loss taxonomy was developed for stationary production equipment. Therefore it does not adequately represent the operational characteristics of autonomous mobile systems operating in dynamic environments.

Within AGVs and AMRs applications, traditional setup and adjustment losses are commonly associated with battery refueling, software updates, vehicle initialization, or task assignment procedures. Idling and minor stoppages are frequently caused by obstacle avoidance, traffic congestion, waiting at intersections, or temporary interactions with human operators. Reduced speed losses arise when vehicles intentionally decrease their speed due to safety constraints, congestion, or navigation requirements. Likewise, process defects are reinterpreted as failed missions, misplaced loads, delivery inaccuracies, or navigation errors requiring manual intervention or mission repetition.

These adaptations demonstrate that although the philosophy of loss identification remains applicable, the nature of the losses changes significantly when moving from fixed manufacturing equipment to autonomous mobile systems. Consequently, several studies emphasize the need for specialized loss taxonomies that accurately represent the operational behavior of autonomous logistics vehicles.

Table 1.3: Adaptation of Nakajima’s Six Big Losses to Mobile Logistics

Classical Six Big Loss	Mobile Logistics Interpretation
Equipment failures	Vehicle hardware or software failures
Setup and adjustment	Battery refueling, software updates, vehicle initialization, task assignment
Idling and minor stops	Obstacle avoidance, traffic congestion, waiting at intersections or doors
Reduced speed	Speed reduction due to safety constraints, congestion, or navigation
Process defects	Failed missions, misplaced loads, delivery errors, navigation failures
Startup losses	Localization, calibration, or system initialization errors

1.6.4 Limitations of OEE for Autonomous Systems

Despite these adaptations, applying classical OEE directly to autonomous systems remains challenging, particularly in outdoor logistics environments. Autonomous vehicles differ from conventional manufacturing equipment in several ways, which complicates the interpretation of the traditional OEE factors.

First, autonomous vehicles operate in changing environments affected by traffic, weather, pedestrians, and interactions with other vehicles. Unlike production machines operating under controlled conditions, variations in mission duration may result from environmental conditions rather than equipment inefficiency.

Second, the fixed-cycle assumption of classical OEE is difficult to apply. Mission durations vary with travel distance, route complexity, traffic, obstacle avoidance, and operational priorities. As a result, defining a single ideal cycle time as a meaningful performance benchmark is difficult.

Finally, autonomous systems generate operational events that have no direct equivalent in conventional manufacturing. Human interventions, remote-operator takeovers, disengagements, and autonomous decision-making failures represent additional loss categories that the traditional OEE framework does not adequately capture. These events affect operational effectiveness and must therefore be considered when adapting OEE to autonomous vehicles.

1.7 Performance Measurement of Autonomous Vehicles

Evaluating autonomous vehicles requires a broader perspective than traditional indicators such as speed, fuel consumption, and travel distance. In industrial environments, performance depends on the vehicle's ability to execute assigned missions safely, reliably, and efficiently.

Unlike conventional vehicles, autonomous systems depend on the interaction between perception, decision-making, navigation, and environmental conditions. Evaluation therefore requires several frameworks and KPIs covering autonomy, safety, reliability, and operational effectiveness.

1.7.1 Levels of Driving Automation

The SAE J3016 standard defines six levels of driving automation, ranging from Level 0 (no automation) to Level 5 (full automation). These levels describe how driving tasks are distributed between the human driver and the automated driving system.

Level 4 automation is particularly relevant to autonomous yard logistics. At this level, the Automated Driving System (ADS) performs all driving tasks without human intervention within a defined ODD. Level 4 systems are therefore suitable for controlled industrial environments such as logistics yards, where operating conditions can be constrained and monitored.

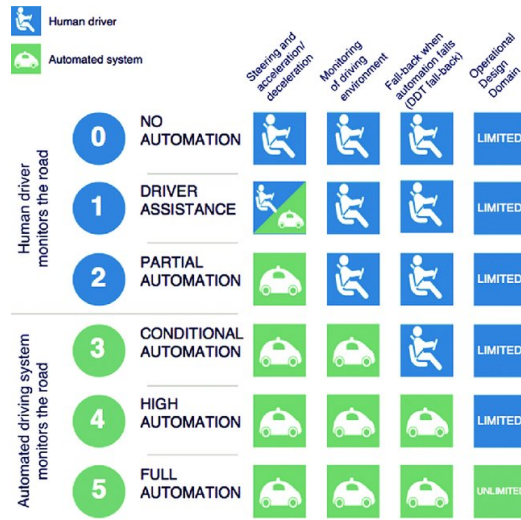


Figure 1.12: SAE Automation Levels [9]

1.7.2 Autonomous-Vehicle Performance Metrics

Autonomous vehicles are commonly evaluated using multiple KPIs, each addressing a specific aspect of system behaviour. Unlike manufacturing systems, which often use OEE as an integrated measure of effectiveness, autonomous-vehicle research generally uses separate metrics for autonomy, safety, reliability, and operational efficiency.

Existing review studies indicate that no universally accepted framework comprehensively evaluates autonomous-vehicle performance. Most available metrics focus on individual dimensions rather than overall operational effectiveness. The main metrics used in this context are summarized in Table 1.4.

Table 1.4: Common Performance Metrics for Autonomous Vehicles

KPI	Description	Performance Aspect
Mission Success Rate	Percentage of missions completed successfully without failure	Mission effectiveness
Intervention / Disengagement Rate	Frequency of human intervention or autonomous disengagement	Autonomy and reliability
Technical Availability	Percentage of time the vehicle is operationally ready	System readiness
Mission Cycle Time	Time required to complete a transport mission	Operational efficiency
Fleet Utilization	Percentage of vehicles actively performing missions	Asset utilization

1.7.3 Operational Design Domain

The ODD defines the conditions under which an ADS is designed to operate safely and effectively. It describes the limits of the system’s operational capability. These limits depend on environmental and operational factors. They also determine when autonomous operation can continue without human intervention.

For autonomous yard vehicles, the ODD specifies the areas in which the vehicle can operate. It also defines the conditions required for autonomous operation. These conditions include the infrastructure, road layout, weather, lighting, traffic, and presence of dynamic obstacles. They may also include speed restrictions, visibility requirements, and interaction with pedestrians or manually operated vehicles.

1.7.4 ODD, SOTIF, and Performance Measurement

ISO 34503 provides a structured approach for describing the ODD of automated driving systems. It organizes ODD characteristics into categories such as scenery, environmental conditions, and dynamic elements [10].

Figure 1.13 illustrates these ODD components. Together, they define the context in which an automated driving system must operate safely. A change in any of these components may affect the vehicle’s ability to continue autonomous operation.

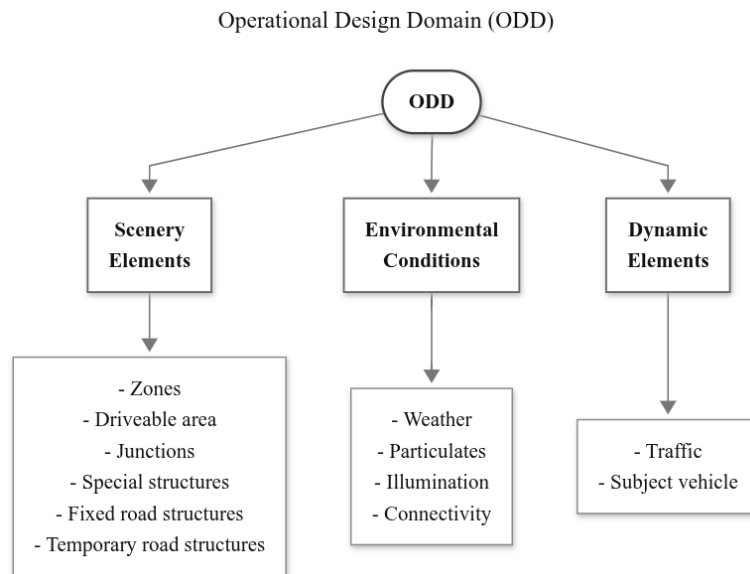


Figure 1.13: Operational Design Domain Components [10]

The Safety of the Intended Functionality (SOTIF) concept, defined by ISO 21448, addresses safety risks caused by functional limitations of an automated system rather than hardware failures [11]. This concept is important for autonomous vehicles operating in complex environments, where unexpected situations may occur even when the hardware remains functional.

For example, fog or other conditions that reduce visibility may prevent an autonomous vehicle from continuing safely despite the correct operation of its hardware components.

The distinction between ODD exclusions and technical failures is therefore important. A technical failure occurs when a system component does not operate as intended, such as in the case of a sensor malfunction, communication failure, or software error. An ODD exclusion occurs when the vehicle encounters conditions outside its defined operational boundaries, such as heavy rain, dense fog, or insufficient visibility. In this case, the vehicle may remain technically functional but unable to continue autonomous operation.

This distinction affects the adaptation of OEE to autonomous vehicles. In conventional manufacturing, Availability losses are mainly associated with equipment breakdowns, maintenance, and setup activities. Autonomous vehicles introduce additional sources of unavailability related to environmental and operational constraints. ODD exclusions should therefore be treated as a distinct category of operational loss when evaluating the Availability factor.

1.8 Complementary Performance Indicators

Although OEE provides a holistic measure of operational effectiveness through the integration of Availability, Performance, and Quality, it does not fully characterize all aspects of autonomous system behaviour. In particular, reliability engineering metrics provide insight into failure behaviour and maintainability, while statistical quality methods allow the evaluation of operational consistency and variability.

1.8.1 Reliability and Maintainability Metrics

Reliability engineering provides established methods for evaluating the robustness and maintainability of complex systems. For autonomous vehicles, these metrics are particularly relevant because operational effectiveness depends not only on task execution but also on the ability of the system to remain available and recover from failures.

The main reliability indicators include:

- **Failure Rate (λ):** represents the frequency at which failures occur during operation. For systems with an approximately constant failure rate, it is commonly related to Mean Time Between Failures (MTBF) as:

$$\lambda = \frac{1}{\text{MTBF}}$$

- **MTBF:** represents the average operating time between two consecutive failures. A higher MTBF indicates improved reliability and fewer interruptions caused by system faults.
- **Mean Time To Repair (MTTR):** represents the average time required to restore the system after a failure. A lower MTTR indicates improved maintainability and faster recovery.

These reliability indicators can be combined to estimate technical availability:

$$A = \frac{\text{MTBF}}{\text{MTBF} + \text{MTTR}}$$

This availability definition describes the proportion of time that a system is operational from a reliability perspective. It directly relates to the Availability component of OEE; however, the two concepts are not identical. Reliability-based availability focuses on the technical state of the system, while OEE Availability considers whether the asset is available for productive operation within the planned production or operational period.

1.8.2 Process Capability

Derived from the framework of Statistical Process Control (SPC), process capability indices C_p and C_{pk} are statistical measures used to evaluate how well a process operates within predefined specification limits [12]. For a process characteristic with lower and upper specification limits (Lower Specification Limit (LSL) and Upper Specification Limit (USL)), process mean μ , and standard deviation σ , the capability indices are defined as:

$$C_p = \frac{\text{USL} - \text{LSL}}{6\sigma}$$

$$C_{pk} = \min\left(\frac{\text{USL} - \mu}{3\sigma}, \frac{\mu - \text{LSL}}{3\sigma}\right)$$

The C_p index represents the potential capability of a process by evaluating the ratio between the allowed specification range and the natural process variability. However, it assumes that the process is perfectly centred. The C_{pk} index additionally considers the deviation of the process mean from the target value and therefore provides a more realistic evaluation of the actual process capability.

In the context of autonomous yard logistics, capability analysis provides complementary information to OEE by addressing the variability of execution performance. While OEE Performance evaluates the loss between actual and ideal execution rates, C_p and C_{pk} quantify the consistency with which repeated operations remain within an expected execution-time range. Therefore, capability analysis directly relates to the fixed-cycle assumption discussed earlier in this chapter, as it evaluates how closely actual mission durations follow the nominal cycle time. This aspect is particularly relevant for autonomous systems, where execution times may vary due to traffic interactions, obstacle avoidance, and dynamic operational conditions.

1.9 Research Gap and Project Positioning

The literature review shows that existing approaches provide valuable tools for evaluating industrial and autonomous systems; however, several limitations remain when applying them to autonomous yard logistics.

Classical OEE was developed for stable manufacturing environments with repetitive operations and predefined cycle times. Its direct application to autonomous vehicles is challenging due to variable mission durations caused by traffic interactions, obstacle avoidance, environmental conditions, and operational constraints. Therefore, the **fixed-cycle assumption underlying the traditional Performance factor requires adaptation**.

Furthermore, autonomous vehicle evaluation relies on multiple KPIs, such as mission success rate, intervention rate, technical availability, and fleet utilization. While these metrics assess specific aspects of operation, they are generally considered independently and do not provide an integrated measure combining availability, performance, and quality losses.

Autonomous systems also introduce new sources of operational unavailability. In particular, **ODD-related limitations must be distinguished from technical failures**, as a vehicle may remain functional but unable to operate under certain environmental conditions. Such losses are not explicitly represented in classical OEE frameworks.

Additionally, reliability metrics (MTBF, MTTR) and capability indices (C_p and C_{pk}) provide complementary information regarding system robustness and execution-time consistency, but they are rarely integrated with OEE for autonomous logistics applications.

1.10 Project Methodology and Work Plan

1.10.1 Development Approach

The internship followed a structured yet flexible approach inspired by Agile principles. Rather than relying on a rigid predefined plan, the work progressed incrementally through data analysis, KPI definition, implementation, testing, and validation. Each stage was reviewed and refined based on practical findings and technical feedback.

The first steps focused on understanding the OEE methodology, system architecture, and available data. The work then progressed toward data preparation, KPI computation, validation, and reporting. Regular exchanges with the company supervisor and updates to the academic supervisor ensured continuous alignment with the project objectives.

1.10.2 Project Timeline

The internship spanned a total of six months, from March 2026 to August 2026. The project was divided into clear, consecutive phases to ensure efficient progress and a logical development workflow. The following timeline outlines the major milestones and activities:

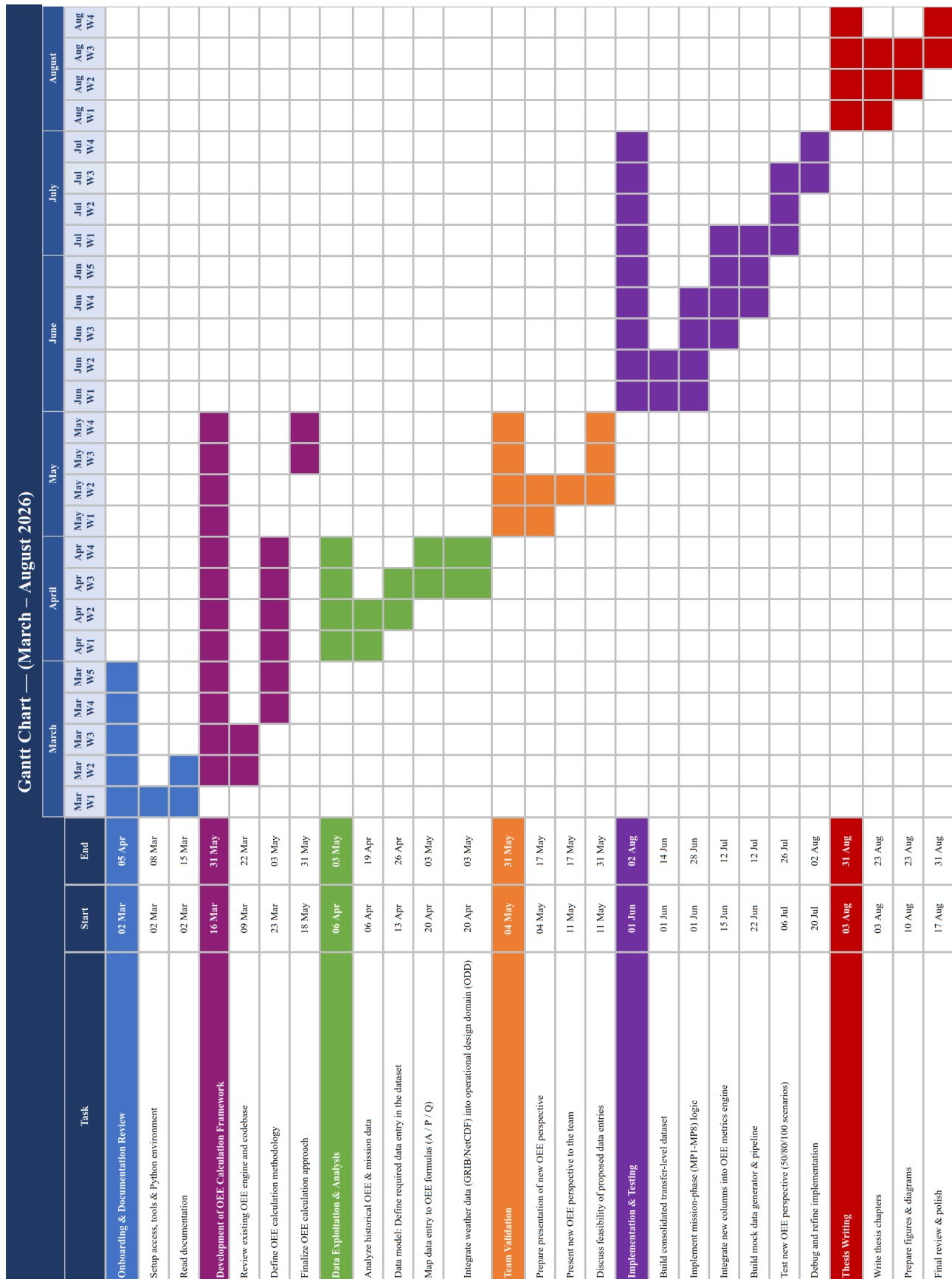


Figure 1.14: Gantt Diagram

1.11 Conclusion

This chapter established the industrial and theoretical foundations of the work. It presented TII KAMAG, the autonomous yard-logistics system, its operating environment, system actors, transfer workflow, mission hierarchy, architecture, and data sources. It also showed that operational effectiveness depends on technical performance, mission variability, human supervision, traffic, and environmental conditions.

The review of OEE showed that Availability, Performance, and Quality provide a useful structure for measuring industrial effectiveness. However, classical OEE was developed for stable manufacturing systems with fixed cycle times. Autonomous yard vehicles operate under variable conditions and generate specific events, such as interventions, disengagements, autonomy limitations, and ODD-related exclusions. Existing autonomous-vehicle KPIs, reliability metrics, and capability indices provide valuable information, but they do not combine these aspects into one integrated measure.

These findings define the research gap addressed by this work. **An adapted OEE framework is required for autonomous yard logistics.** The framework must account for variable mission durations, autonomy-specific losses, supervision requirements, and ODD-related constraints. The following chapter presents its measurement unit, time hierarchy, adapted OEE factors, loss classification, ODD treatment, and complementary indicators.

Chapter 2

Design and Data Architecture of the Proposed OEE Framework

2.1 Introduction

The previous chapter identified several limitations in applying classical OEE to autonomous yard logistics. These limitations include variable mission durations, heterogeneous interruptions, human supervision, and ODD-related restrictions. Classical OEE does not explicitly represent these characteristics. Its concepts must therefore be adapted to produce meaningful results for autonomous transport operations.

This chapter presents the OEE-based framework developed for autonomous yard logistics. It preserves the classical relationship between Availability, Performance, and Quality. It also adapts the measurement unit, time hierarchy, loss classification, and interpretation of these factors. The main contributions include a **transfer-based measurement unit**, a **six-level time hierarchy**, an **ODD-aware time-accounting method**, and a **decomposition of Performance into dispatch utilisation and execution speed**.

The chapter also introduces a transfer-outcome taxonomy and complementary indicators for reliability, supervision, throughput, environmental influence, and process capability. It assesses the availability of the industrial data required for implementation and defines a canonical data model for structuring this information. These elements provide the basis for the implementation and validation presented in the following chapter.

2.2 Design Principles of the Adapted OEE Framework

Classical OEE was developed to evaluate fixed manufacturing assets. It measures effectiveness through equipment availability, operating speed, and product quality. Applying it directly to autonomous yard-logistics vehicles is not straightforward because their operating conditions differ from those of conventional production systems. In this work, I preserve the mathematical structure of classical OEE while adapting its concepts to autonomous transport operations.

I developed the framework around five main design decisions. First, I selected the transfer

as the main unit of productive output, while retaining missions and mission phases for diagnostic analysis. Second, I introduced scenario- and phase-dependent nominal durations to account for the variability of autonomous transport operations. Third, I decomposed Performance into dispatch utilisation and execution speed. Fourth, I separated ODD-related exclusions from technical downtime through a sequential time hierarchy and overlap-correction rules. Finally, I distinguished transfer outcomes from autonomy and supervision indicators. **These decisions adapt OEE to the variable and constrained nature of autonomous transport operations.**

Based on these decisions, I maintain the relationship $OEE = A \times P \times Q$ while adapting the interpretation of each factor to autonomous yard logistics. I also ensure that operating time and operational losses are attributed completely and without overlap. The resulting framework remains mathematically consistent while providing detailed information about autonomous system performance.

2.3 The OEE Measurement Unit

I selected the transfer as the primary OEE measurement unit because it represents the complete logistics service delivered by the vehicle. In classical OEE, the productive output is usually represented by a manufactured product. In autonomous yard logistics, however, the system provides a transport service rather than producing a physical product. The transfer therefore serves as the natural accounting unit. It represents one complete transport order that moves a swap body from a source location to a destination location.

As introduced in Section 1.3, the operational process has three levels: transfer, mission, and mission phase. I evaluate OEE primarily at the transfer level, while retaining missions and mission phases for diagnostic analysis. This structure preserves the complete operational context. It also allows losses identified at transfer level to be traced to the underlying execution sequence. Figure 2.1 illustrates this measurement hierarchy.

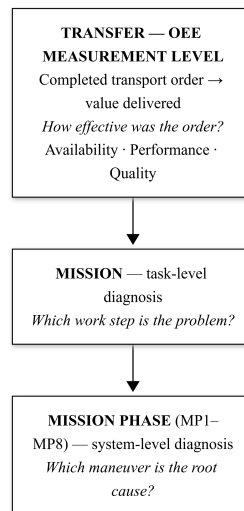


Figure 2.1: Hierarchical Organisation of the Operational Process

2.4 The Six-Level Time Hierarchy

I developed a six-level time hierarchy to separate scheduling losses, planned exclusions, ODD restrictions, unplanned downtime, performance losses, and quality losses. Each level is obtained from the previous one by subtracting a specific category of loss or exclusion. This sequential structure assigns every part of the operating period to one category. It also prevents double counting and keeps the OEE factors mathematically consistent. The overlap-correction mechanism for ODD exclusions and downtime is detailed in Section 3.3.

The hierarchy starts with total calendar time T_t . This represents the complete evaluation period. I remove non-operating periods, such as nights, weekends, and scheduled non-operating days, to obtain Planned Operating Time T_o .

I then remove planned stoppages, such as maintenance, refuelling, and system preparation. I also remove periods during which operation is not expected because the vehicle is outside its Operational Design Domain. The remaining time is Required Time T_r . Unplanned interruptions during Required Time are then removed to obtain Available Time T_a .

Next, I subtract performance losses to obtain Performance-Adjusted Time T_n . This value represents the nominal-speed equivalent duration of all qualifying started transfers. Finally, I subtract quality losses to obtain Useful Time T_u . This value represents the nominal-speed equivalent duration of successfully completed transfers.

Table 2.1 defines the six time levels.

Table 2.1: Definition of the Six-Level Time Hierarchy

Term	Symbol	Definition
Calendar Time	T_t	Complete evaluation horizon (e.g., $24 \text{ h} \times 365 \text{ d}$), used as the absolute time reference
Planned Operating Time	T_o	Scheduled operating window after removing non-operating periods
Required Time	T_r	T_o minus planned and ODD-related exclusion
Available Time	T_a	T_r minus unplanned downtime
Performance-Adjusted Time	T_n	Nominal-speed equivalent time of all qualifying started transfers
Useful Time	T_u	Nominal-speed equivalent time of successfully completed transfers

Figure 2.2 illustrates the complete time hierarchy.

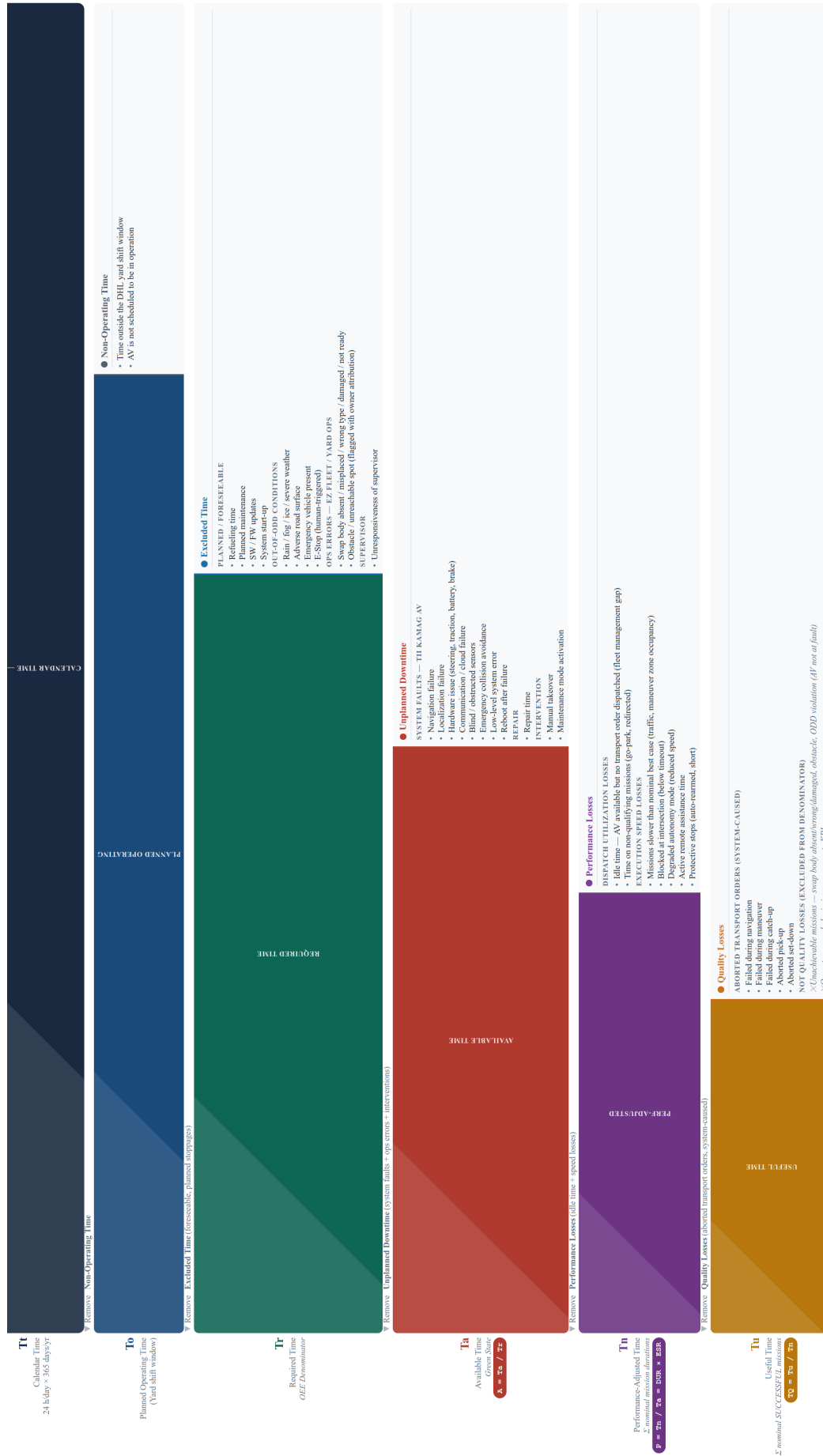


Figure 2.2: Six-Level Time Hierarchy of the Proposed OEE Framework

The resulting decomposition is expressed as:

$$T_t \xrightarrow{\text{-non-operating}} T_o \xrightarrow{\text{-ODD exclusions}} T_r \xrightarrow{\text{-unplanned downtime}} T_a \xrightarrow{\text{-performance losses}} T_n \xrightarrow{\text{-quality losses}} T_u$$

Based on this hierarchy, I define the three OEE components as:

$$A = \frac{T_a}{T_r}, \quad P = \frac{T_n}{T_a}, \quad Q = \frac{T_u}{T_n},$$

Therefore:

$$\text{OEE} = A \times P \times Q = \frac{T_u}{T_r}.$$

The proposed hierarchy differs from classical OEE in two important ways. First, Required Time T_r excludes ODD-related periods. The vehicle is therefore not penalised for conditions under which autonomous operation is not permitted. Second, I calculate Performance and Quality using the nominal durations of qualifying and successful transfers. This preserves the relationship $\text{OEE} = T_u/T_r$ and maintains mathematical consistency.

2.5 Availability

Availability measures the fraction of **Required Time** during which the autonomous vehicle is operationally available:

$$A = \frac{T_a}{T_r} = \frac{T_r - \text{Unplanned Downtime}}{T_r}.$$

I defined Required Time so that planned stoppages and ODD-related periods are removed before Availability is calculated. Since T_r already excludes planned stoppages and periods outside the Operational Design Domain (ODD) (Section 3.3), Availability reflects exclusively **unplanned downtime**. Consequently, ODD-related exclusions reduce the OEE denominator but do not affect the Availability factor. It ensures that the vehicle is not penalised for conditions under which autonomous operation is not permitted.

Unplanned downtime events are classified according to three complementary dimensions, as illustrated in Figure 2.3.

- **Category** describes the nature of the interruption. Two principal categories are distinguished:
 - **System faults**, corresponding to failures originating from the autonomous system itself, such as localisation failures, trajectory deviations, or low-level system errors.
 - **Human interventions**, corresponding to operator-initiated interruptions, including manual vehicle takeover and client-requested transfer aborts.

- **Owner** identifies the responsible party (autonomous system or operations), enabling consistent attribution of losses and supporting root-cause analysis.
- **Failure status** distinguishes engineering failures from supervisory or operational events, providing the basis for the complementary reliability metrics presented in Section 3.4.3.

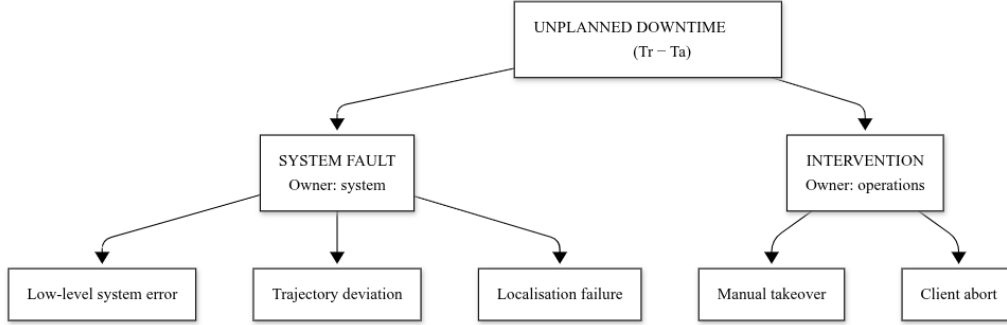


Figure 2.3: Classification of Unplanned Downtime Events

2.6 Performance

I decomposed Performance into the Dispatch Utilisation Rate (DUR) and the Execution Speed Rate (ESR). This separation distinguishes unused Available Time from delays during mission execution.

Classical OEE evaluates Performance by comparing actual production speed with an ideal cycle time. This approach is not sufficient for autonomous yard logistics. Low Performance may result from two different causes. The vehicle may remain available without executing a mission. Alternatively, an assigned mission may take longer than its nominal duration.

To distinguish these causes, I define Performance as the product of two factors:

$$P = \text{DUR} \times \text{ESR}.$$

T_e denote the total actual execution time of qualifying started transfers. Let T_n denote the sum of their corresponding nominal durations.

I define the **Dispatch Utilisation Rate** as:

$$\text{DUR} = \frac{T_e}{T_a}.$$

DUR measures the proportion of Available Time spent executing qualifying transport operations. It captures losses caused by a lack of orders, dispatch delays, waiting between tasks, and underutilisation of the vehicle.

I define the **Execution Speed Rate** as:

$$\text{ESR} = \frac{T_n}{T_e}.$$

ESR compares the nominal duration of the executed work with its actual duration. It captures losses caused by slow navigation, waiting during execution, obstacle interactions, retries, and other delays that extend mission duration.

The decomposition preserves the original Performance factor:

$$P = \frac{T_e}{T_a} \times \frac{T_n}{T_e} = \frac{T_n}{T_a}.$$

This decomposition provides more diagnostic information than a single Performance ratio. A low DUR indicates that the available vehicle was not sufficiently utilised. A low ESR indicates that assigned work was executed more slowly than its nominal reference. These two causes can therefore be analysed separately.

Finally, scenario- and phase-dependent nominal durations reduce the limitations of the fixed-cycle assumption. Each transfer is compared with an appropriate operational reference rather than with one common cycle time.

2.7 Quality

I introduced a five-class transfer-outcome taxonomy to represent successful, manually assisted, aborted, cancelled, and unachievable operations.

In autonomous yard logistics, quality cannot be assessed through the conformity of a manufactured product. Instead, it is defined by the outcome of the transport operation. The proposed framework therefore introduces a five-class outcome taxonomy that distinguishes successful, unsuccessful, and excluded transport executions.

Table 2.2: Outcome Taxonomy for Quality Assessment in Autonomous Yard Logistics

Outcome	Description	Contribution
Completed Au- tonomous	Transfer completed without human intervention	Successful
Completed Manual	Transfer completed following manual intervention	Quality Loss
Aborted	Transfer started but not completed	Quality loss
Cancelled	Transfer cancelled before execution	Excluded
Unachievable	Transfer impossible because of operational constraints	Excluded

Transfers completed autonomously deliver the required transport service and are therefore considered successful. Aborted transfers and manually completed transfers consume operational resources without producing the intended outcome and consequently constitute quality losses.

Cancelled and unachievable transfers are excluded because execution was never expected under valid operating conditions.

To capture different aspects of operational quality, two complementary indicators are defined.

The **time-based quality**

$$Q_{\text{time}} = \frac{T_u}{T_n}$$

is used in the computation of OEE and preserves the fundamental relationship

$$\text{OEE} = A \times P \times Q = \frac{T_u}{T_r}.$$

The **count-based quality**

$$Q_{\text{count}} = \frac{N_{\text{successful}}}{N_{\text{started}}}$$

provides an operational view of mission success independent of execution duration. Reporting both indicators distinguishes losses caused by unsuccessful executions from those resulting from differences in mission duration or route complexity.

2.8 The OEE family — OEE, GEE, and TEEP

Three nested indicators are defined using the same useful operating time T_u while varying only the time horizon considered: Global Equipment Effectiveness (GEE) and Total Effective Equipment Performance (TEEP) alongside OEE.

Table 2.3: Comparison of OEE, GEE, and TEEP

Metric	Formula	Interpretation
OEE	$\frac{T_u}{T_r}$	How effectively the available operational time was converted into useful output
GEE	$\frac{T_u}{T_o}$	How effectively the planned operating window was exploited
TEEP	$\frac{T_u}{T_t}$	How effectively total calendar time was exploited

Because the corresponding denominators are nested:

$$T_t \geq T_o \geq T_r$$

the following relationship always holds:

$$\text{TEEP} = \frac{T_u}{T_t} \leq \text{GEE} = \frac{T_u}{T_o} \leq \text{OEE} = \frac{T_u}{T_r}$$

The differences between these indicators provide additional diagnostic information. The gap between GEE and OEE reflects the exclusions applied between Planned Operating Time T_o and Required Time T_r , including planned stoppages and ODD-related exclusions within the operating window. The gap between TEEP and GEE reflects the calendar time outside the planned operating schedule, such as nights, weekends, and other non-operating periods.

2.9 ODD Exclusion Framework — Weather ODD, Operational ODD, and Double-Counting Prevention

I designed the ODD-aware time-accounting method to prevent environmental and operational restrictions from being classified as technical failures.

The Operational Design Domain (ODD), introduced in Section 2.2, defines the environmental and operational conditions under which the autonomous driving system is expected to perform its intended function. Periods in which the vehicle operates outside its ODD do not represent equipment unavailability; rather, operation is not required or permitted under those conditions. Therefore, ODD-excluded periods are removed from the Required Time T_r definition instead of being treated as Availability losses.

Within the proposed framework, ODD exclusion is implemented by subtracting the corresponding excluded duration from the transition between Planned Operating Time T_o and Required Time T_r . The framework distinguishes two complementary sources of ODD exclusion: weather-related constraints and operational constraints.

2.9.1 Weather ODD

Weather-based exclusions represent environmental conditions that exceed the operational limits defined for the autonomous vehicle. Hourly meteorological observations are evaluated against configurable thresholds stored in the `ODD_Parameters` table.

Table 2.4: Weather-Based ODD Exclusion Criteria

Condition	Parameter	Exclusion criterion
Heavy rain	<code>precip_threshold_mm_h</code>	Precipitation rate above threshold
Snowfall	<code>snow_threshold_mm_h</code>	Water-equivalent snowfall rate above threshold
Strong wind	<code>wind_threshold_ms</code>	Wind speed above threshold
Freezing conditions	<code>temp_freeze_c</code>	Temperature below threshold
Fog	<code>fog_dewpoint_diff_k</code> + <code>cloud_cover</code>	Dew-point depression below threshold with cloud coverage above 80%

Each shift hour violating at least one weather constraint is classified as ODD-excluded, and its duration is stored, thereby reducing the Required Time T_r . When multiple weather

constraints occur simultaneously, a fixed precedence rule (rain > snow > wind > freezing > fog) is applied only for event attribution and traceability.

2.9.2 Operational ODD

In addition to environmental constraints, autonomous yard operations are affected by external operational conditions that determine whether a valid mission can be executed. Examples include an unavailable swap body, a blocked destination position, or an incorrect swap body type assignment. These conditions prevent mission execution independently of the vehicle capability.

Each operational ODD exclusion is classified similarly, and the accumulated duration is stored after applying weather-based exclusions, with the total daily ODD exclusion capped at the planned shift duration.

The Required Time is therefore calculated as:

$$T_r = T_o - T_{startup} - T_{refueling} - T_{SW/FW} - T_{planned_maint} - T_{out_of_odd}$$

where $T_{out_of_odd}$ represents the total duration excluded due to weather and operational ODD violations.

2.9.3 Double-Counting Prevention

I introduced overlap-correction rules to ensure that each period is assigned to only one exclusion or loss category.

ODD periods may overlap with downtime or with other operational exclusions. Without correction, the same duration could be removed more than once.

For example, a technical interruption may occur during an hour that has already been excluded because of weather. Deducting its full duration from Available Time would penalise the vehicle for a period during which operation was not required.

For each event, only the part outside an already excluded ODD period is retained:

$$T_{effective} = T_{event} - T_{event \cap ODD}$$

or equivalently:

$$T_{effective} = T_{event}(1 - f_{overlap})$$

where $f_{overlap}$ represents the fraction of the event duration overlapping with an already excluded ODD period.

This correction is applied to both unplanned downtime events and operational ODD events during the transfer-level time-hierarchy computation (Section 3.4.1), ensuring that every lost second is attributed to exactly one category within the time hierarchy. Consequently, ODD

limitations affect the operational time requirement without artificially reducing Availability, preserving the physical interpretation of the OEE factors.

2.10 Operational Deployment Indicators

The proposed framework defines complementary indicators to explain the technical and operational causes behind the aggregate OEE value. These indicators cover reliability, execution robustness, human supervision, operational capacity, environmental influence, and execution-time capability. Where the required input data are available, they can be computed from the same canonical data structure used for OEE calculation.

2.10.1 Engineering Reliability Metrics — MTBF, MTTR, and Failure Rate

As introduced in Section 1.8.1, MTBF, MTTR, and failure rate provide complementary information on system reliability and maintainability. Within the proposed framework, these indicators are used to distinguish the technical failure behaviour of the autonomous vehicle from the broader operational losses captured by OEE.

Only genuine technical or equipment failures are considered reliability events. Supervisory interventions, operational constraints, and ODD-related exclusions are excluded because they do not directly represent hardware or software failures.

For the proposed framework, MTBF is calculated using Available Time T_a as the operational exposure time:

$$\text{MTBF} = \frac{T_a}{N_{\text{fail}}}$$

while MTTR represents the average repair or recovery duration associated with the detected failures:

$$\text{MTTR} = \frac{\sum_{i=1}^{N_{\text{fail}}} t_{\text{recovery},i}}{N_{\text{fail}}}.$$

The corresponding failure rate is obtained as defined in Section 1.8.1:

$$\lambda = \frac{1}{\text{MTBF}}.$$

These indicators complement OEE Availability by distinguishing whether technical losses result primarily from frequent failures or from prolonged recovery periods:

Table 2.5: Reliability Metric Definitions

Metric	Definition	Purpose
MTBF	Average operational time between failures	Measures failure frequency
MTTR	Average recovery duration after failure	Measures maintainability
λ	Failure rate	Represents failure occurrence intensity

2.10.2 Process-quality indicators

The Quality factor of OEE evaluates whether a transfer is successfully completed; however, successful completion alone does not fully represent the maturity of an autonomous system. A vehicle may maintain a stable Quality value while requiring increasing human support, additional execution attempts, or more frequent recovery actions.

To identify such gradual degradation, process-quality indicators are calculated at the transfer level. These indicators act as leading measures of autonomous capability by monitoring execution behaviour before it results in measurable OEE losses.

Table 2.6: Process-Quality Indicators

Indicator	Definition
First-attempt pickup rate	Ratio of transfers completed successfully during the first pickup attempt to the total number of qualifying pickup operations
First-attempt set-down rate	Ratio of transfers completed successfully during the first set-down attempt to the total number of qualifying set-down operations
Retry rate	Ratio of transfers requiring at least one additional execution attempt to the total number of qualifying transfers
Remote validation rate	Ratio of transfers requiring at least one remote validation to the total number of qualifying transfers
Protective-stop rate	Ratio of transfers experiencing at least one protective stop to the total number of qualifying transfers

These indicators provide insight into the robustness of autonomous execution and reveal increasing dependence on supervision that may remain hidden in aggregated OEE values.

2.10.3 Human supervision and autonomy indicators

Autonomous yard systems are designed to reduce manual involvement; however, supervision remains an important operational factor during deployment. Therefore, indicators related to

human assistance are monitored:

Table 2.7: Human Supervision and Autonomy Indicators

Indicator	Definition
Intervention rate	Number of remote interventions per completed transfer
Autonomous completion ratio	Proportion of transfers completed without human intervention
Remote assistance duration	Total time spent requiring remote operator support

These indicators quantify the remaining human workload and allow evaluation of the progression toward higher levels of autonomous operation.

2.10.4 Capacity and throughput indicators

OEE measures the effectiveness of required operating time but does not directly represent the achieved logistics capacity. Therefore, throughput-oriented indicators are included to evaluate the operational output of the autonomous fleet.

Table 2.8: Capacity and Throughput Indicators

Indicator	Definition
Transfers per operating day	Number of completed transfers during the defined operating window
Transfers per operating hour	Average transfer throughput during active operation
Transfer duration distribution	Statistical distribution of transfer execution times across scenarios

These indicators connect the effectiveness measurement framework with practical logistics performance requirements.

2.10.5 Environmental influence indicators

Since autonomous vehicle performance depends on environmental and operational conditions, ODD-related information is analysed together with operational outcomes.

Table 2.9: Environmental Influence Indicators

Indicator	Definition
ODD exclusion duration	Total time removed from Required Time due to weather or operational ODD limitations
Weather versus intervention correlation	Relationship between environmental conditions and human intervention frequency
Weather versus abort correlation	Relationship between environmental conditions and unsuccessful mission outcomes

These indicators support the identification of external factors affecting autonomous operation and help distinguish vehicle limitations from environmental constraints.

2.10.6 Execution-time capability analysis

In addition to execution success and operational throughput, the consistency of autonomous execution is evaluated through process capability analysis. While process-quality indicators describe **how** operations are executed, capability analysis evaluates **how consistently** execution duration remains within an expected range.

Execution-time capability is evaluated using transfer and mission durations relative to a tolerance band centred around the nominal duration:

$$USL = 1.3 \times T_{\text{nominal}}$$

$$LSL = 0.7 \times T_{\text{nominal}}$$

The capability indices are calculated as:

$$C_p = \frac{USL - LSL}{6\sigma}$$

$$C_{pk} = \min \left(\frac{USL - \mu}{3\sigma}, \frac{\mu - LSL}{3\sigma} \right)$$

where μ and σ represent the mean and standard deviation of the measured execution durations.

The C_p index evaluates the spread of execution times relative to the defined tolerance interval, whereas C_{pk} additionally considers the deviation of the process mean from the nominal duration. Therefore, C_{pk} can identify systematic slowdowns or accelerations even when execution variability remains limited.

Capability analysis is performed at two levels:

- **Transfer-scenario level:** evaluates end-to-end predictability from an operational and logistics perspective.
- **Mission-phase level:** evaluates the consistency of individual MP1–MP8 execution durations relative to their phase-specific nominal references.

Together, these complementary indicators extend the OEE framework from a pure effectiveness measurement toward a broader evaluation of autonomous system maturity, operational capacity, and deployment readiness.

2.11 Industrial Data Assessment and OEE Input Availability

I assessed the datasets used to implement the proposed framework and evaluated their suitability for OEE computation. This section presents the available industrial and environmental data sources. It then examines their temporal coverage, granularity, consistency, and completeness. Finally, it compares the information required by the OEE model with the information available in the source records.

This assessment has two objectives. First, it identifies the data constraints that motivate the processing and reconstruction methods used in the implementation pipeline. Second, it identifies the additional fields required for more reliable OEE computation in future deployments. I therefore use the real data to assess the feasibility of the proposed approach and define its data requirements. The data is not treated as a complete or controlled measurement campaign.

2.11.1 Available Industrial Data Sources

The implementation uses two operational datasets: an automatic dataset export and a manual field log. These sources differ in structure, granularity, and content. The automatic dataset provides machine-generated information, while the field log adds operator-recorded context.

2.11.1.1 Automatic dataset

The automatic dataset covers two acquisition campaigns. Phase 1 ran from **11 December 2022 to 10 February 2023**. Phase 2 took place in **March 2023**. The dataset contains three linked levels:

- 425 transfers with 99 attributes;
- 750 missions with approximately 38 attributes;
- 3,137 incidents with approximately 28 attributes.

Transfer records describe complete transport orders. They include retry, abort, error, soft-stop, disengagement, and manual-mode counters. They also contain operating-mode durations, travelled distances, and GPS information.

Mission records include the mission type, duration, distance, state, and error information. Incident records capture events such as soft stops, protective stops, disengagements, and transitions to manual operation.

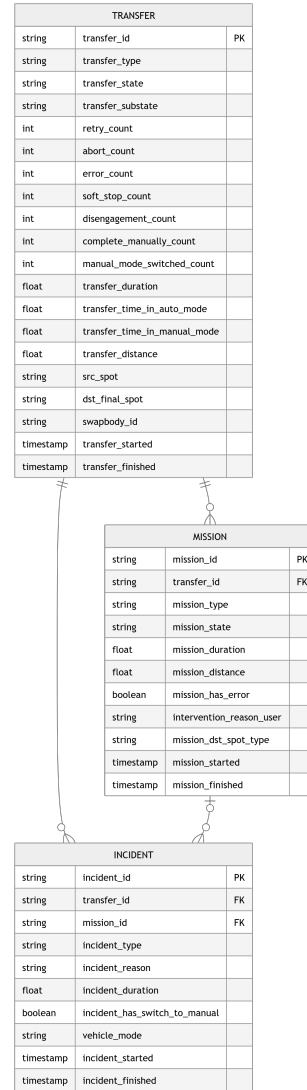


Figure 2.4: Automatic dataset source data model

2.11.1.2 Manual field-log dataset

The manual dataset contains **944** operator-recorded mission attempts across seven test weeks. Phase 1 includes **weeks 50, 2, 3, 5, and 6**. Phase 2 includes **weeks 10 and 11**. Each row represents one mission attempt and contains approximately 25 attributes.

These attributes cover identification, timing, outcome, human intervention, and environmental context. They include mission and scenario information, start and completion times, manoeuvre and catch-up durations, success or failure status, failure location and reason, manual-assistance details, ODD compliance, and free-text comments about weather and operating con-

ditions.

The records are distributed across several weekly worksheets, but they share the same flat structure. Each row represents one observed mission attempt. Unlike the automatic dataset, the manual log does not separate transfers, missions, and incidents into linked levels. It also lacks common identifiers such as `transfer_id` and `mission_id`.

The two sources therefore cannot be merged through a direct identifier-based join. I instead identify corresponding records using temporal and operational attributes, including the execution date, timestamp, mission type, and source and destination positions.

MISSION_ATTEMPT	
date	log_date
string	mode
string	site
string	safety_chain
string	scenario
string	starting_area
string	type
string	starting_id
string	destination_id
time	starting_time
time	maneuver_zone_entry_time
time	end_of_mission_time
string	success_failed
string	if_failed_where
string	if_failed_reason
string	odd_respected
string	manual_intervention
string	intervention_where
string	intervention_reason
time	catch_up_time
time	maneuver_time
time	mission_time
string	weather_comment

Figure 2.5: Manual field-log source data model

2.11.2 Availability of Required OEE Inputs

The datasets were not collected as part of a dedicated OEE measurement campaign. The automatic and manual sources cover partially overlapping periods. They also differ in structure, identifiers, and temporal precision. The automatic dataset contains system-generated transfer and mission identifiers. The manual log contains independent operator observations. I therefore harmonise and reconcile the two sources before combining them.

Several inputs required by the OEE framework are not recorded explicitly. These include planned operating schedules, planned activity durations, nominal execution times, structured downtime categories, and complete repair information. Some quantities can be reconstructed from existing fields. Others require documented configuration values or engineering assumptions.

Table 2.10 summarises the readiness of the available data for OEE computation. The classification distinguishes between directly available information, values derived from source

records, and inputs that must be configured or estimated.

Table 2.10: Availability of source-data inputs for OEE computation

OEE element	Required information	Availability in the source data
Calendar Time T_t	Observation-period boundaries	Directly available
Planned Operating Time T_o	Shift calendar and operating window	Not recorded
Required Time T_r	Planned activities and ODD exclusions	Partially available
Available Time T_a	Downtime duration and classification	Partially available
Performance-Adjusted Time T_n	Actual and nominal execution durations	Actual durations available; nominal durations absent
Useful Time T_u	Standardised transfer outcomes	Derivable from existing fields
Reliability indicators	Failure and repair information	Incomplete
Process-quality indicators	Retries, interventions, and stop events	Distributed across both sources

2.12 Canonical Data Model

2.12.1 Purpose and Design Objectives

The automatic dataset and the manual field log presented in Section 2.11.2 differ in structure, granularity, identifiers, and information content. The automatic source organises information across transfers, missions, and incidents, whereas the manual source records independent mission observations without shared system identifiers. Moreover, neither source contains all the information required by the proposed OEE framework in a directly usable form.

I therefore developed a canonical data model to provide a unified representation of the required information. The model does not reproduce either source dataset. Instead, it defines a common target structure. The available automatic and manual records are transformed into this structure before being processed by the metrics engine. The canonical model serves two purposes. In the present implementation, it provides a standard input structure for consolidating the heterogeneous industrial records. It also defines a recommended structure for future data collection within the company. Future systems can record OEE-relevant information directly in this form, reducing the need for reconstruction and manual interpretation.

As shown in Figure 2.6, the core model consists of six operational entities: **Transfers**, **Missions**, **Downtime_Events**, **Speed_Loss_Events**, **Out_of_ODD_Events**, and **Shift_Schedule**.

The **Transfers** entity represents the central operational unit, while the supporting entities store information defined at a finer or broader level of granularity.

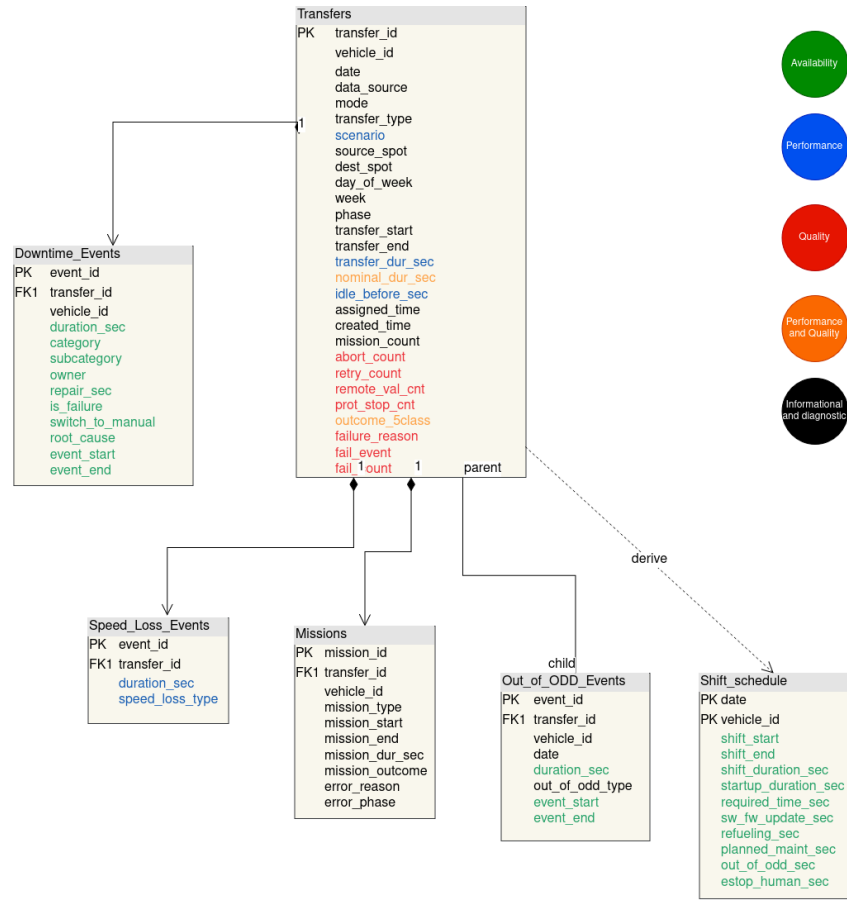


Figure 2.6: Canonical entity-relationship model and mapping of attributes to the OEE dimensions

2.12.2 Transfer as the Central Operational Entity

The **Transfers** entity forms the centre of the canonical model. A transfer represents a complete transport order performed between a source position and a destination position. It provides a meaningful unit of analysis because each transfer has a defined execution interval, an operational scenario, a final outcome, and a nominal-duration reference.

Transfer records provide the main inputs for Performance and Quality. Actual and nominal durations are used to evaluate execution speed, while the standardised outcome determines whether the transfer contributes to Useful Time. Availability is calculated separately at vehicle-and-operating-day level using **Shift_Schedule** and **Downtime_Events**. The transfer-level results are then aggregated over the same evaluation period to obtain the complete OEE.

The fields contained in **Transfers** can be grouped according to their analytical role, as summarised in Table 2.11. Only the principal fields are highlighted here; the complete set is shown in Figure 2.6.

Table 2.11: Main information groups in the Transfers entity

Information group	Representative fields	Purpose
Identification and context	<code>transfer_id</code> , <code>vehicle_id</code> , <code>date</code> , <code>data_source</code> , <code>mode</code>	Identify the operation and preserve its source and operating context
Route and scenario	<code>source_spot</code> , <code>dest_spot</code> , <code>transfer_type</code> , <code>scenario</code>	Describe the transfer configuration and select the appropriate nominal reference
Timing	<code>transfer_start</code> , <code>transfer_end</code> , <code>transfer_dur_sec</code> , <code>nominal_dur_sec</code>	Support execution-time and Performance calculations
Execution	<code>mission_count</code> , <code>abort_count</code> , <code>retry_count</code>	Describe how the transfer was completed
Outcome	<code>outcome_5class</code> , <code>failure_reason</code>	Support Quality and failure analysis
Diagnostic indicators	<code>remote_val_cnt</code> , <code>prot_stop_cnt</code> , <code>1st_pickup</code> , <code>1st_setdown</code>	Support supervision, reliability, and process-quality analysis

The standardised outcome taxonomy distinguishes between `COMPLETED_AUTO`, `COMPLETED_MANUAL`, `ABORTED`, `CANCELED`, and `UNACHIEVABLE`. This classification ensures that transfers originating from different source datasets are interpreted consistently by the metrics engine.

2.12.3 Supporting Entities and Relationships

The supporting entities represent information recorded at a different level from the complete transfer. `Missions` preserves the operational tasks composing each transfer, while `Downtime_Events`, `Speed_Loss_Events`, and `Out_of_ODD_Events` represent distinct types of operational losses or exclusions. `Shift_Schedule` stores the planned operating information defined for each vehicle and operating day.

Each mission is linked to one transfer through `transfer_id`. Downtime and speed-loss events may also be linked to a specific transfer. However, this relationship can remain optional because some events may occur between transfers. Similarly, an ODD event may affect a specific transfer or a broader operating interval. It can therefore also be associated with the vehicle and operating date.

`Shift_Schedule` is independent of individual transfers because a scheduled operating period exists even when no transfer is executed. It is identified by the combination of `date` and `vehicle_id`. Table 2.12 summarises the level of granularity and analytical role of each entity.

Table 2.12: Entities and levels of granularity in the canonical data model

Entity	Level of granularity	Principal key	Main analytical role
Transfers	One complete transport order	<code>transfer_id</code>	Performance and Quality analysis
Missions	One operational task within a transfer	<code>mission_id</code>	Sub-transfer diagnostics
Downtime_Events	One unplanned interruption	<code>event_id</code>	Availability and reliability
Speed_Loss_Events	One non-stopping execution loss	<code>event_id</code>	Performance-loss diagnosis
Out_of_ODD_Events	One ODD-exclusion interval	<code>event_id</code>	Required-Time determination
Shift_Schedule	One vehicle and operating day	<code>date</code> , <code>vehicle_id</code>	Planned Operating Time and Required Time

2.12.4 Mapping to the OEE Dimensions

The colour coding used in Figure 2.6 indicates the OEE dimension supported by each field. Green fields contribute primarily to Availability, blue fields to Performance, red fields to Quality, and orange fields to both Performance and Quality. Black fields provide diagnostic information.

Table 2.13: Mapping of the OEE time hierarchy to the canonical data model

OEE quantity	Symbol	Principal source	Implementation
Calendar Time	T_t	Analysis configuration	Duration of the selected calendar evaluation period
Planned Operating Time	T_o	Shift_Schedule	Sum of <code>shift_duration_sec</code> over scheduled operating periods
Required Time	T_r	Shift_Schedule, Out_of_ODD_Events	Planned Operating Time after planned and ODD-related exclusions
Available Time	T_a	Downtime_Events	Required Time minus qualifying unplanned downtime
Performance-Adjusted Time	T_n	Transfers	Sum of <code>nominal_dur_sec</code> for qualifying executed transfers
Useful Time	T_u	Transfers	Sum of <code>nominal_dur_sec</code> for transfers satisfying the useful-outcome criteria

2.12.5 Role as a Reference Data Structure

I discussed the proposed canonical model with members of the engineering team. The discussion focused on its potential use for future data collection and framework development. The model identifies the information that should be recorded systematically to support OEE computation.

This information includes common identifiers, shift schedules, event timestamps, nominal durations, failure categories, repair times, and ODD exclusions. I therefore present the model as a technical recommendation. It does not represent an already deployed company data architecture.

In this work, I first populate the model using the available automatic and manual records. This step tests whether the existing industrial data can be transformed into the proposed structure. It also identifies which fields are directly available, which can be reconstructed, and which require estimates or configuration assumptions.

I then test the model using controlled synthetic data. These datasets contain complete and predefined inputs. They make it possible to evaluate the data structure and calculation engine under known operating conditions and loss scenarios.

The two datasets serve different purposes. The industrial data demonstrates the feasibility of applying the proposed model to existing records. The synthetic data supports controlled validation of its completeness, consistency, and suitability for OEE computation.

2.13 Framework Summary

The proposed framework extends the classical OEE methodology to the evaluation of autonomous yard logistics systems while preserving the original structure. The principal adaptations introduced in this work include the definition of the transfer as the measurement unit, the extension of the time hierarchy to account for autonomous operation, the explicit treatment of Operational Design Domain (ODD) exclusions, and the introduction of autonomous-vehicle-specific loss classifications. Figure 2.7 presents an overview of the adopted framework.

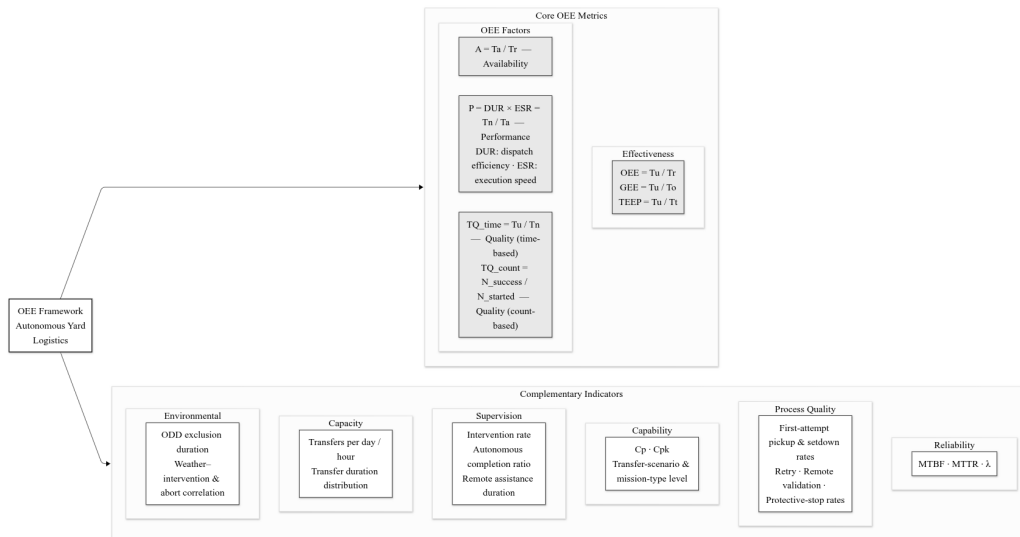


Figure 2.7: Overview of the proposed OEE-based performance evaluation framework

Availability, Performance, and Quality remain the primary measures used to compute OEE, GEE, and TEEP, whereas complementary indicators support diagnosis of operational performance without altering the OEE calculation itself.

Reliability metrics (MTBF, MTTR, and failure rate) characterise the technical behaviour of the autonomous vehicle, while process-quality indicators monitor the robustness of mission execution. Process capability indices (Cp and Cpk) evaluate the consistency of execution times, and operational indicators such as intervention rate, supervision workload, throughput, and weather correlation provide additional information regarding deployment readiness.

The framework definition is complemented by an assessment of the available industrial inputs and by the specification of a canonical data model. Together, these additions connect the conceptual OEE structure to the data architecture required for implementation.

2.14 Conclusion

In this chapter, I developed an OEE-based framework for autonomous yard logistics. I retained the classical relationship between Availability, Performance, and Quality. I then adapted the quantities used to calculate these factors to the operational characteristics of autonomous transport.

I selected the transfer as the primary measurement unit and developed a six-level time hierarchy. I also decomposed Performance into dispatch utilisation and execution speed. This decomposition separates vehicle underuse from delays during mission execution. In addition, I introduced an ODD-aware time-accounting method that distinguishes external operating restrictions from technical unavailability.

I defined overlap-correction rules and a transfer-outcome taxonomy for autonomous operations. I also incorporated complementary indicators for reliability, human supervision, throughput, environmental influence, process quality, and execution-time capability. Together, these indicators provide additional diagnostic information without changing the core OEE calculation.

Finally, I assessed the availability of the industrial inputs required by the framework. I identified the main gaps in the source data and defined a canonical data model containing transfers, missions, events, ODD exclusions, and shift information. This model aligns the available data with the proposed time hierarchy and OEE factors.

These contributions adapt classical OEE to the operational characteristics of autonomous yard logistics. **The resulting framework links the conceptual OEE model to the data structure required for implementation and validation.** The following chapter presents the implementation, validation, and evaluation of the framework. It covers data integration, event classification, nominal-duration estimation, ODD integration, KPI computation, and controlled validation.

Chapter 3

Implementation and Validation of the Proposed OEE Framework

3.1 Introduction

The previous chapter defined the proposed OEE framework for autonomous yard logistics. This chapter presents how I implemented and validated the framework. The implementation combines a **transfer-level OEE engine with a mission-phase layer for more detailed operational analysis**.

The implementation uses the canonical data model defined in the previous chapter and applies it to the available industrial records through a structured ingestion pipeline. However, due to incomplete and heterogeneous records, the resulting indicators are treated as illustrative rather than as a definitive evaluation of vehicle performance. **Controlled synthetic scenarios were therefore developed to validate the calculation logic, sensitivity, and consistency of the framework.**

The chapter first presents the implementation components: the data integration and model-ingestion pipeline, weather-data processing for ODD integration, the transfer-level metrics and reporting engine, and the mission-phase diagnostic extension. It then presents the validation approach, the controlled synthetic validation results, and the mission-phase diagnostic validation.

3.2 Data Integration and Model Ingestion Pipeline

The automatic dataset export and manual field log were transformed into the canonical model defined in Section 2.12 through a three-stage pipeline. The process first harmonises and cross-matches the two sources, then reconstructs complete transfers and consolidates the available information, and finally populates the canonical entities and derives the missing OEE inputs. The resulting workflow is illustrated in Figure 3.1.

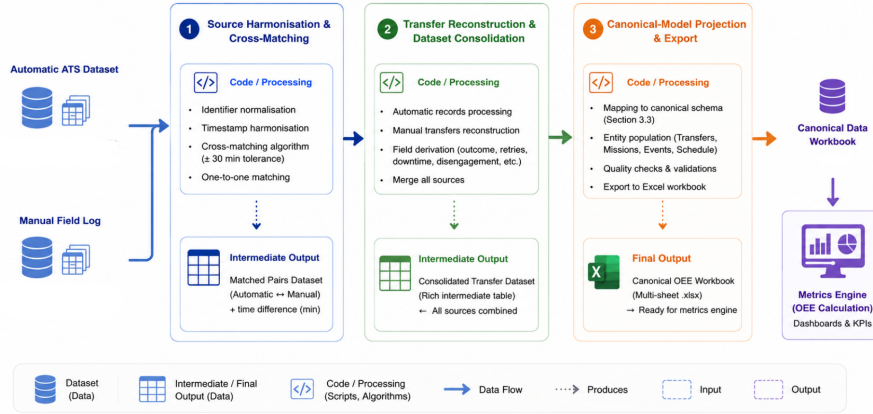


Figure 3.1: End-to-end processing pipeline for canonical-model ingestion

3.2.1 Source Harmonisation and Automatic-Manual Cross-Matching

The automatic and manual datasets use different identifiers, timestamp formats, and levels of precision. Destination identifiers were therefore normalised into a common format, while date and time fields were converted into comparable timestamps.

Records were matched using execution date, mission type, destination position, and start time. Date, mission type, and destination had to match exactly after normalisation, while a maximum start-time difference of 30 min was allowed. A one-to-one rule was applied so that each automatic mission could be associated with only one manual observation. The matching criteria are summarised in Table 3.1.

Table 3.1: Automatic-manual cross-matching rules

Criterion	Automatic source	Manual source	Matching rule
Destination	Normalised destination identifier	Normalised destination identifier	Exact
Execution date	Date extracted from mission timestamp	Recorded date	Exact
Mission type	<code>mission_type</code>	Recorded mission type	Exact after normalisation
Start time	Automatic mission start	Manual starting time	Maximum difference of 30 min

The procedure was applied to **750** automatic missions and **944** manual attempts. It produced 101 matched pairs. This limited number is mainly explained by the partial overlap between the acquisition periods, the difference in timestamp precision, and the absence of a shared mission identifier. The matched dataset preserves the fields from both sources and records the time difference used during the association.

3.2.2 Transfer Reconstruction and Dataset Consolidation

Matched and unmatched records from Section 3.2.1 are reconstructed into transfer-level observations and consolidated into a common intermediate dataset.

For the automatic source, the existing transfer structure is preserved, with missions and incidents linked through the original system identifiers. When a manual match is available, its contextual information is attached to the same operation.

Manual observations require transfer reconstruction because no transfer identifier is available. Consecutive mission attempts are grouped when they occur on the same day and the gap between them does not exceed 15 min. A reconstructed transfer generally consists of a pick-up followed by a set-down, with an optional parking movement.

Matched records enrich overlapping observations, while unmatched records from both sources are retained when they contain usable information. After standardisation of timestamps, positions, operating mode, and data origin, the consolidated dataset contains **908 transfers** and **1,692 missions**, including **101 matched automatic–manual pairs**.

3.2.3 Mapping the Consolidated Data to the Canonical Model

The consolidated transfer and mission datasets are then mapped to the canonical structure defined in Section 2.12. Table 3.2 summarises the population of the canonical entities.

Table 3.2: Population of the canonical entities

Canonical entity	Main input	Population approach
Transfers	Consolidated transfer dataset	Direct column mapping and standardised derived fields
Missions	Consolidated mission dataset	Direct column mapping using <code>transfer_id</code>
Downtime_Events	Error and disengagement information	Conversion into structured unplanned-downtime records
Speed_Loss_Events	Recorded protective-stop duration	One record per transfer with non-zero soft-stop time
Out_of_ODD_Events	Operational failure reasons and ODD conditions	Classification into structured ODD-event records
Shift_Schedule	Operating dates and configured schedule values	One record per vehicle and operating day

Transfers and **Missions** are populated from the corresponding consolidated records, while event-level entities are generated from the available operational information. Error and disengagement records are converted into **Downtime_Events**. Protective-stop durations are instead represented as **Speed_Loss_Events**, since they increase execution time without necessarily representing loss of Availability. This separation prevents the same interval from affecting both Availability and Performance.

Operational ODD conditions are mapped into **Out_of_ODD_Events**, while weather-related ODD events are added separately through the process described in Section 3.3. A **Shift_Schedule**

record is generated for each observed vehicle and operating day.

3.2.4 Derivation and Configuration of Missing OEE Inputs

Several inputs required by the OEE framework are not directly available in the industrial records. I therefore complete the calculation chain using three approaches. Some values are configured from project information. Others are derived from the observed data. When no measurement is available, I use a documented assumption. Table 3.3 summarises these inputs and their role in the implementation.

I define the operating schedule from project documentation. The planned shift runs **from 07:00 to 13:00, Monday to Friday**. Public holidays are excluded from the planned operating period. I also introduce startup and refuelling durations as planned exclusions when calculating Required Time. No software or firmware update events were recorded during the analysed period. I therefore set their duration to zero and assume that they occur outside the operating shift.

Nominal transfer durations require a separate treatment because the source data contains no reference duration. I derive a scenario-specific value from successful transfers completed without errors or retries. For each scenario, I define the nominal duration as 1.20 times the minimum duration observed among these clean transfers:

$$t_{\text{nominal},s} = 1.20 \times \min_{i \in C_s} (t_{\text{actual},i})$$

Here, C_s represents the set of clean successful transfers for scenario s . When the available reference observations are insufficient, I use a fallback nominal value of **120 s**. This fallback affects **37** transfers, corresponding to **4.1%** of the consolidated dataset. I also apply a transfer-duration cap of 7,200 s as an outlier-control rule. This prevents abnormally long transfers from distorting the Execution Speed Rate.

Other quantities also require substitute values. Manual emergency-stop events have no recorded duration. I therefore assign **30 s** to each event. This concerns three downtime events. Repair duration is also missing for part of the dataset. When no measured repair interval is available, I use a value of **5 min** per failure. This affects **268** transfers and is particularly relevant to the manual records.

I apply two additional limits during reconstruction. Recovery gaps are capped at 8 h. This prevents overnight periods from being interpreted as downtime. Manual mission attempts are grouped into the same reconstructed transfer only when the gap between consecutive operations does not exceed **15 min**.

Together, these configured, derived, and fallback values complete the input set required by the canonical model described in Section 2.12.

Table 3.3: Configured and derived inputs used in the implementation

Parameter	Value	Type	Basis	Main use
Shift window	07:00–13:00, Mon–Fri	Configured	Project documentation	Planned Operating Time (T_o)
Public holidays	4 days	Configured	Operating calendar	Planned Operating Time (T_o)
Startup duration	10 min/day	Configured	Project documentation	Required Time (T_r)
Refuelling duration	20 min/fortnight	Configured	Project documentation	Required Time (T_r)
Software/firmware updates	0 min	Configured assumption	No recorded events; assumed outside shift	Required Time (T_r)
Transfer-duration cap	7,200 s	Configured	Outlier control	ESR, T_n
Nominal duration	$1.20 \times \text{min.}$ clean-success duration	Statistically derived	Scenario-specific transfer data	ESR, T_n , T_u
Nominal fallback	120 s	Fallback	Insufficient reference observations (4.1% of transfers, $n = 37$)	ESR, T_n
Manual emergency-stop duration	30 s/event	Fallback	No recorded event duration (0.3% of transfers, $n = 3$)	Available Time (T_a)
Manual repair duration	5 min/failure	Fallback	No measured repair interval (29.5% of transfers, $n = 268$; 55.5% of manual transfers)	MTTR
Recovery-gap cap	8 h	Configured	Excludes overnight intervals	Downtime reconstruction
Manual grouping gap	15 min	Configured	Max. gap within one reconstructed transfer	Transfer reconstruction

3.3 Weather Data Processing and ODD Integration

The ODD-accounting method defined in Section 2.9 requires environmental conditions to be converted into time-based exclusions that can be used by the metrics engine. This section describes how this process was implemented. Weather observations are first prepared at hourly

resolution, then compared with the configured ODD limits. Violating periods are finally converted into `Out_of_ODD_Events` and integrated into the daily operating schedule.

3.3.1 Weather Data Acquisition and Preparation

Meteorological data was obtained from the **ERA5 hourly single-level reanalysis dataset** provided through the Copernicus Climate Data Store [13]. ERA5 combines numerical model information with observations to provide a spatially and temporally consistent representation of atmospheric conditions. The selected product provides hourly data on a global regular latitude–longitude grid with a spatial resolution of $0.25^\circ \times 0.25^\circ$.

ERA5 was used because no local weather station provided continuous measurements for the analysed period. The dataset therefore provides a consistent external source for reconstructing the environmental conditions corresponding to the available operational records.

For the present implementation, the available weather archives cover December 2022 to April 2023. The grid point closest to the operating site was selected at 48.50°N , 10.25°E . The required atmospheric variables were extracted and converted into the units used by the ODD-classification procedure.

Table 3.4: Weather variables used for ODD classification

ERA5 variable	Physical quantity	Transformation	Resulting unit
<code>t2m</code>	2 m air temperature	$T - 273.15$	$^\circ\text{C}$
<code>d2m</code>	2 m dew-point temperature	$T_d - 273.15$	$^\circ\text{C}$
<code>u10, v10</code>	10 m wind components	$\sqrt{u_{10}^2 + v_{10}^2}$	m/s
<code>tp</code>	Total precipitation	$\times 1000$	mm
<code>sf</code>	Snowfall water equivalent	$\times 1000$	mm
<code>tcc</code>	Total cloud cover	$\times 100$	%

The ERA5 source provides hourly estimates of atmospheric and land-surface quantities, which makes it suitable for comparison with the hourly ODD logic implemented in the framework.

The converted weather observations are then aligned with the operating schedule. Only weather records corresponding to a scheduled operating interval are relevant for OEE computation. Conditions outside this period do not generate an ODD exclusion.

3.3.2 Hourly Weather-ODD Classification

Each retained hourly observation is evaluated against the configurable weather limits stored in `ODD_Parameters`. Five environmental conditions are considered: heavy rain, snowfall, strong wind, freezing temperature, and fog.

An hour is classified as outside the weather ODD when at least one of these conditions exceeds its corresponding limit. When several conditions occur simultaneously, the hour is excluded only once. A precedence rule may be used to assign a dominant cause for reporting, but it does not increase the excluded duration.

Table 3.5: Hourly weather-based ODD exclusion criteria

Weather condition	ERA5 variables	Exclusion criterion
Heavy rain	Total precipitation	≥ 5.0 mm/h
Snowfall	Snowfall water equivalent	≥ 2.0 mm/h
Strong wind	10 m wind speed	≥ 15.0 m/s
Freezing conditions	2 m air temperature	≤ -5.0 °C
Fog proxy	2 m air temperature, 2 m dew-point temperature, total cloud cover	$ T - T_d < 2.5$ K and cloud cover $> 80\%$

The fog condition is implemented as a proxy because no direct fog variable is used in the processing pipeline. A small difference between air temperature and dew-point temperature, combined with high cloud cover, is used to identify atmospheric conditions associated with reduced visibility.

The threshold values are stored as configuration parameters rather than embedded directly in the processing logic. They can therefore be modified without changing the weather-processing workflow.

3.3.3 ODD Event Generation and Canonical-Model Integration

After hourly classification, the violating records are aggregated by operating day. The weather-based exclusion time is calculated from the proportion of observed shift hours classified as outside the ODD:

$$T_{\text{ODD,wx}} = \frac{N_{\text{bad}}}{N_{\text{observed}}} T_{\text{shift}},$$

where N_{bad} is the number of violating hourly observations, N_{observed} is the number of available weather observations within the corresponding operating period, and T_{shift} is the scheduled shift duration.

The proportional calculation allows the procedure to remain applicable when the available weather records do not cover the complete operating window.

This integration connects the weather-processing module directly to the canonical model defined in Section 2.12. The metrics engine therefore receives the ODD exclusion through the same standardised structure as the other OEE inputs. Its treatment within the $T_o \rightarrow T_r$ transition follows the methodology already defined in Section 2.9.

Figure 3.2 illustrates the weather-ODD processing and canonical-model integration workflow.

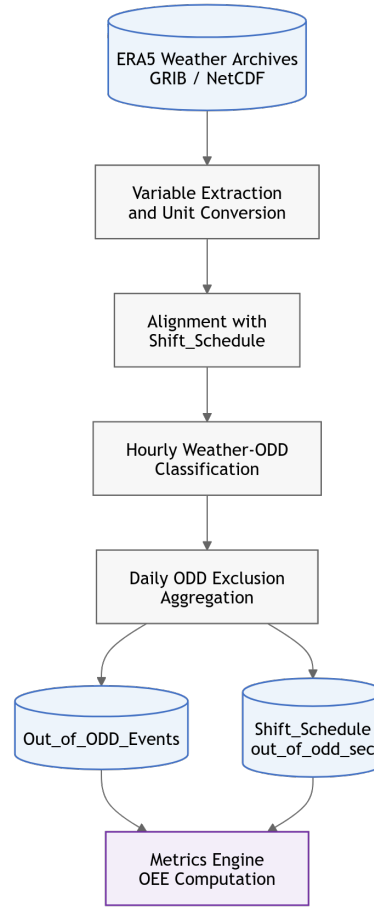


Figure 3.2: Weather-ODD processing and canonical-model integration workflow

3.4 Transfer-Level Metrics and Reporting Engine

The transfer-level metrics engine is the analytical core of the framework. It takes the canonical dataset produced by the integration pipeline (Section 3.2), enriched with ODD information (Section 3.3), and applies the calculation rules from Chapter 2 to derive the six-level time hierarchy, the OEE factors, and the complementary operational indicators. This section presents the resulting analysis based on the consolidated industrial dataset.

3.4.1 Time-Hierarchy Computation

The engine computes the six time levels sequentially:

$$T_t \rightarrow T_o \rightarrow T_r \rightarrow T_a \rightarrow T_n \rightarrow T_u.$$

Their mathematical definitions were established in Section 2.4 and are therefore not repeated here. The implementation applies these definitions directly to the consolidated dataset.

The analysed period covers **95 calendar days**, corresponding to **2,280.00 h** of Calendar

Time. The populated shift schedule contains 44 vehicle-shift entries, representing **264.00 h** of Planned Operating Time. After the configured planned exclusions and ODD-related periods are removed, **139.40 h** remain as Required Time.

The effective unplanned downtime is subsequently deducted to obtain 79.56 h of Available Time. The nominal durations associated with qualifying started transfers produce 17.20 h of Performance-Adjusted Time. Finally, the nominal contribution of autonomously completed transfers results in 10.16 h of Useful Time. These values are consistent with the time hierarchy used by the current metrics implementation.

Table 3.6: Consolidated OEE time hierarchy

Time level	Symbol	Value
Calendar Time	T_t	2,280.00 h
Planned Operating Time	T_o	264.00 h
Required Time	T_r	139.40 h
Available Time	T_a	79.56 h
Performance-Adjusted Time	T_n	17.20 h
Useful Time	T_u	10.16 h

3.4.2 OEE Factor Computation

The time hierarchy is used to calculate the three OEE factors. For the consolidated dataset, Availability is:

$$A = \frac{T_a}{T_r} = \frac{79.56}{139.40} = 57.08\%.$$

Performance is:

$$P = \frac{T_n}{T_a} = \frac{17.20}{79.56} = 21.62\%.$$

The time-based Quality factor is:

$$Q = \frac{T_u}{T_n} = \frac{10.16}{17.20} = 59.08\%.$$

The resulting Overall Equipment Effectiveness is:

$$\text{OEE} = \frac{T_u}{T_r} = 7.29\%.$$

The factor formulation provides the corresponding internal consistency check:

$$\text{OEE} = A \times P \times Q = 57.08\% \times 21.62\% \times 59.08\% = 7.29\%.$$

The engine also reports a count-based Quality indicator. For the consolidated dataset, this

value is 60.41%. The difference from the time-based value arises because the nominal duration varies between transfer scenarios; consequently, a transfer does not necessarily contribute the same amount of nominal time as another.

Performance Decomposition

Performance is further decomposed into the Dispatch Utilisation Rate and Execution Speed Rate introduced in Section 2.6. For the consolidated dataset, the updated calculation gives:

Table 3.7: Performance decomposition of the consolidated dataset

Indicator	Value
DUR	100.00%
ESR	21.62%
Performance	21.62%

The decomposition therefore satisfies:

$$P = \text{DUR} \times \text{ESR} = 100.00\% \times 21.62\% = 21.62\%.$$

The DUR reaches its maximum value in the consolidated evaluation. Consequently, the observed Performance loss is represented primarily by the ESR rather than by additional dispatch-utilisation loss. An ESR of 21.62% indicates a substantial difference between the nominal execution reference and the observed transfer durations.

A DUR of 100% should not be interpreted as continuous vehicle motion. Transfer duration may contain protective stops, waiting periods, remote assistance, and other execution delays. These losses remain part of the observed transfer duration and therefore affect execution speed.

3.4.3 Reliability and Process-Quality Indicators

In addition to the headline OEE factors, the metrics engine generates complementary indicators to support the diagnosis of operational losses. Reliability-related analysis focuses on the recorded unplanned downtime, while process-quality indicators describe recurring execution characteristics that are not directly visible in the global OEE value.

For the engineering reliability analysis, only events classified as technical `system_fault` events were retained as failures. Operational interruptions such as supervisory interventions, disengagements, operational errors, and ODD-related exclusions were not included in the failure count. For the consolidated dataset, 338 technical failure events were identified over 79.563 h of Available Time.

Table 3.8: Engineering reliability indicators for the consolidated industrial dataset

Indicator	Value
Technical failure events	338
Available Time T_a	79.563 h
Total repair/recovery duration	130.4 min
MTBF	0.235 h
MTTR	0.39 min
Failure rate λ	4.248 failures/h

In addition to failure frequency, the distribution of unplanned downtime was analysed to identify the main contributors to lost Available Time. The recorded downtime events were ranked according to their accumulated duration, as presented in Table 3.9.

Table 3.9: Pareto analysis of recorded unplanned downtime

Rank	Downtime category	Number of events	Recorded duration (h)	Share of downtime (%)	Cumulative share (%)
1	System fault	142	107.213	69.73	69.73
2	Recovery gap	209	36.927	24.02	93.75
3	Intervention	3	5.879	3.82	97.57
4	Disengagement	120	2.726	1.77	99.34
5	Operational error	16	1.019	0.66	100.00
Total	—	490	153.764	100.00	—

System faults constitute the dominant category, accounting for 69.73% of the recorded downtime duration. Recovery gaps contribute a further 24.02%. Together, these two categories represent 93.75% of the total recorded downtime, indicating that the distribution is strongly concentrated in a small number of loss categories.

The durations shown in the Pareto analysis correspond to the recorded or reconstructed downtime events. They are used primarily for loss diagnosis and prioritisation. The effective downtime used in the Availability calculation may be lower after ODD-overlap correction and time-hierarchy constraints are applied.

Process-quality indicators provide a complementary view of transfer execution. They capture retries, first-attempt success, remote assistance, and protective stops that may affect operational performance without necessarily representing technical downtime.

The results show that pickup operations have a lower first-attempt success rate than set-down operations. Retries occur in approximately one fifth of qualifying transfers, while protective stops affect more than one quarter of them. Remote validation is required less frequently but remains relevant for evaluating the level of human supervision.

Table 3.10: Process-quality indicators for the consolidated dataset

Indicator	Value
First-Attempt Pickup Rate	67.76%
First-Attempt Set-down Rate	84.30%
Retry Rate	22.44%
Remote Validation Rate	10.82%
Protective Stop Rate	28.42%

3.4.4 Execution-Time Distribution and Capability

The Performance factor relies on scenario-specific nominal durations. The variability of the observed transfer durations was therefore examined before evaluating execution-time capability.

Figure 3.3 shows the distribution of transfer durations in the consolidated industrial dataset. Most transfers are concentrated between 2 and 20 minutes, while a smaller proportion exhibits considerably longer execution times. This long-duration tail indicates substantial variability in the operational data and helps explain the difference between actual and nominal execution times.

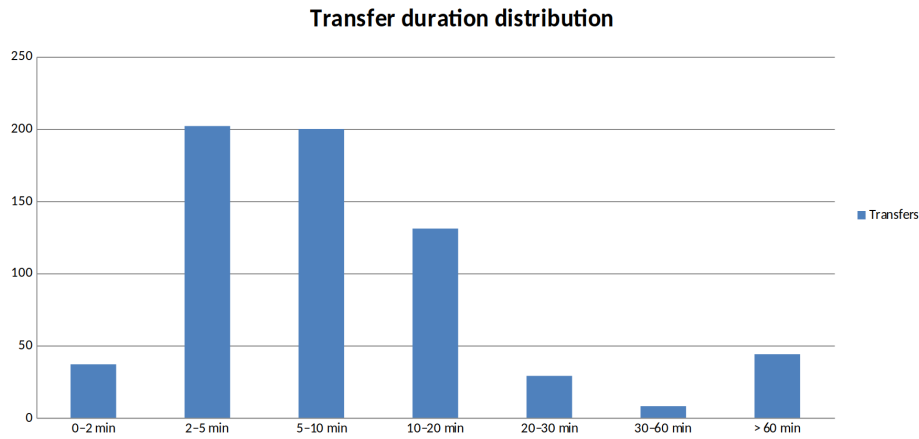


Figure 3.3: Distribution of transfer durations in the consolidated industrial dataset

To complement this analysis, execution-time capability was evaluated for each transfer scenario. These scenarios include park, close-proximity parking, and Road Map File (RMF) parking configurations. The engine compares the actual duration of each eligible transfer with its scenario-specific nominal duration and calculates the C_p and C_{pk} indices using the tolerance limits defined in Section 2.10.6.

The results are presented in Table 3.11. All evaluated scenarios show very low C_p values and negative C_{pk} values. The mean actual-to-nominal duration ratios range from approximately 4.75 to 20.71, indicating that the observed execution times are substantially higher than their corresponding nominal references.

Table 3.11: Transfer execution-time capability by scenario

Scenario	n	Nominal duration (s)	Mean duration (s)	Mean actual/nominal ratio	C_p	C_{pk}	Interpretation
Close-proximity parking	13	77	1,595.0	20.71	0.0029	-0.1906	Not capable
Park	89	32	519.2	16.23	0.0025	-0.1246	Not capable
RMF parking	14	77	1,031.1	13.39	0.0044	-0.1770	Not capable
Unclassified	14	77	366.0	4.75	0.0285	-0.3279	Not capable

The negative C_{pk} values indicate that the mean execution time lies outside the selected tolerance range. The process is therefore not capable relative to the current nominal-duration references. This result is consistent with the low ESR obtained in Section 3.4.2 and highlights the difference between the derived best-case nominal durations and the execution times observed under industrial operating conditions.

3.4.5 OEE Family and Reporting Outputs

The metrics engine also calculates the broader OEE-family indicators introduced in Section 2.8. For the consolidated dataset:

$$\text{GEE} = \frac{T_u}{T_o} = 3.85\%,$$

and

$$\text{TEEP} = \frac{T_u}{T_t} = 0.45\%.$$

These indicators use progressively broader time references than OEE. GEE relates Useful Time to the planned operating period, whereas TEEP relates it to the complete calendar evaluation horizon. They are reported alongside OEE to make the effect of scheduling and planned exclusions visible without changing the headline OEE definition.

The final Excel workbook contains both summary and diagnostic outputs. In addition to the consolidated OEE results, it provides daily and weekly breakdowns, downtime and speed-loss analyses, capability results, process-quality indicators, statistical monitoring outputs, and data-lineage information. The reporting layer therefore separates the headline effectiveness value from the detailed indicators required for engineering diagnosis.

Only the principal industrial-data outputs are presented in this section. The framework supports a broader set of complementary indicators, but their evaluation depends on the completeness of the corresponding input data. Indicators that could not be evaluated reliably from the available records are retained as part of the framework specification and are discussed in the limitations and future data requirements.

3.5 Mission-Phase Diagnostic Extension

The transfer-level analysis in Section 3.4 provides the authoritative OEE result. However, it does not show which mission phase causes a delay, retry, or other loss.

A mission-phase diagnostic extension was therefore added using the MP1–MP8 sequence defined in Table 1.1.

The industrial records do not contain reliable timestamps for each mission phase. The phase-level dataset is therefore reconstructed synthetically from the observed transfer and mission records.

The resulting indicators are used for diagnostic demonstration only. They do not represent directly measured phase-level performance. The transfer-level OEE remains unchanged.

3.5.1 Mission-Phase Data Structure

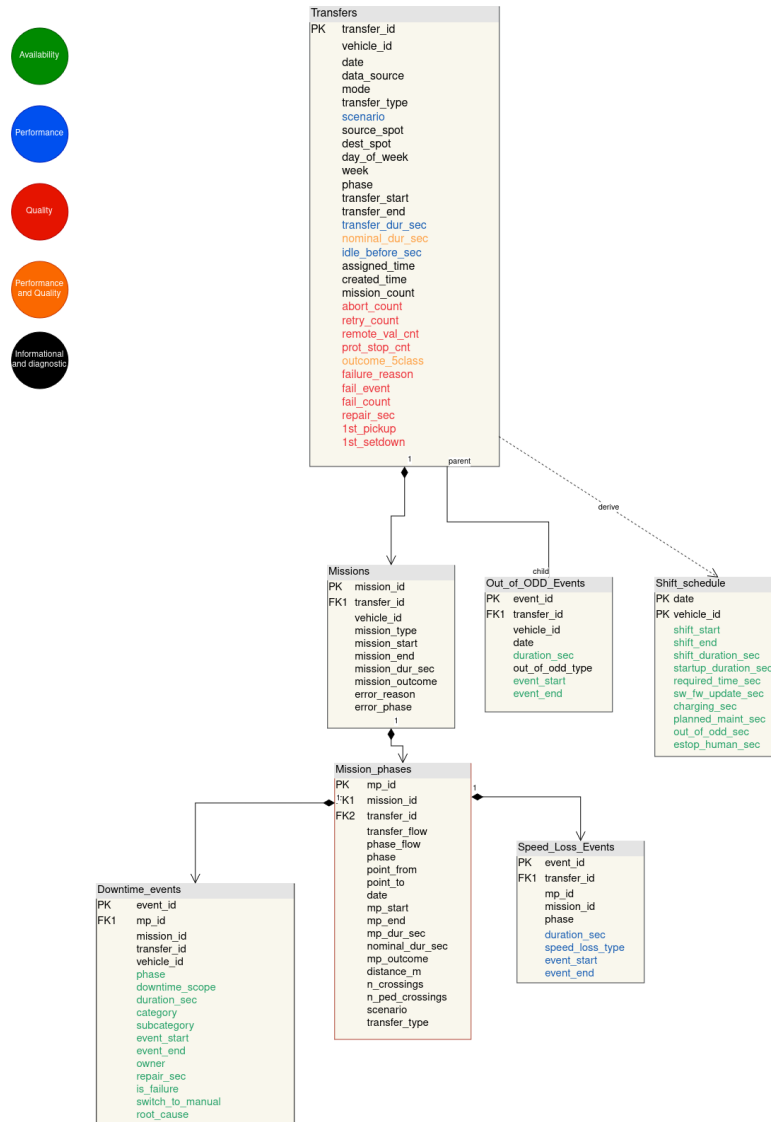


Figure 3.4: Data model of the mission-phase diagnostic extension

The canonical model was extended with a `Mission_phases` entity. This creates an additional level between missions and phase-specific diagnostic events. The resulting hierarchy is:

Transfer → Mission → Mission Phase.

Each **Mission_phases** record represents one execution instance of one phase. It remains linked to its parent mission and transfer. The entity contains the phase sequence, reconstructed timing, actual duration, nominal duration, outcome, and operational context.

For MP1, additional route information is stored. This includes travelled distance and the number of road and pedestrian crossings. These fields are required because the MP1 nominal duration depends on the travelled path.

Downtime and speed-loss records can also be linked to a phase through **mp_id**. This link is optional. Events that cannot be assigned reliably to one phase remain associated with the mission or transfer level. Table 3.12 summarises the main information groups in the **Mission_phases** entity and their role in the diagnostic analysis.

Table 3.12: Main information groups in the **Mission_phases** entity

Information group	Main attributes	Purpose
Identification	mp_id , mission_id , transfer_id	Preserve the transfer–mission–phase hierarchy
Sequence	transfer_flow , phase_flow , phase	Locate the phase within the mission
Timing	mp_start , mp_end , mp_dur_sec	Represent reconstructed execution
Nominal reference	nominal_dur_sec	Support ESR and capability analysis
Outcome	mp_outcome	Support phase-level Quality analysis
Route context	distance_m , n_crossings , n_ped_crossings	Support MP1 nominal-duration calculation
Operational context	scenario , transfer_type	Support diagnostic grouping

3.5.2 Mission-Phase Reference and Nominal Durations

After defining the structure, a nominal-duration reference is assigned to each phase.

The phase-level nominal durations use a different method from the transfer-level nominal durations defined in Section 3.2.4. Transfer-level values are statistically derived from the industrial records. Phase-level values are configured specifically for the controlled diagnostic layer.

MP2–MP8 use fixed reference durations. These values are modelling assumptions in the current implementation. MP1 uses a variable duration because path-following time depends on the route.

Table 3.13 summarises the nominal-duration configuration used for the mission-phase layer. The nominal duration of MP1 is calculated as:

Table 3.13: Mission-phase nominal-duration configuration

Phase	Description	Nominal duration	Basis
MP1	Follow predefined path	Distance-dependent	Calculated
MP2	Manoeuvre toward pre-insertion point	45 s	Configured assumption
MP3	Insert under swap body	10 s	Configured assumption
MP4	Pick up swap body	5 s	Configured assumption
MP5	Insert into empty spot	10 s	Configured assumption
MP6	Set down swap body	3 s	Configured assumption
MP7	Extract from spot	10 s	Configured assumption
MP8	Catch up predefined path	20 s	Configured assumption

$$T_{\text{nom},\text{MP1}} = \frac{d}{v} + 5n_c + 5n_p,$$

where d is the travelled distance, $v = 15 \text{ km/h}$ is the nominal path-following speed, n_c is the number of road crossings, and n_p is the number of pedestrian crossings. A 5 s allowance is configured for each crossing.

3.5.3 Construction of the Synthetic Mission-Phase Dataset

The industrial logs contain transfer and mission durations but do not contain phase boundaries. The `Mission_phases` records must therefore be reconstructed.

For each eligible transfer, the recorded scenario is first identified. The corresponding mission and phase sequence is then obtained from the reference configuration. The MP1–MP8 sequences were introduced earlier in Table 1.1 and are therefore not repeated here. A nominal duration is first assigned to each phase using the rules defined in Section 3.5.2. However, these nominal values represent reference execution times and do not reproduce the duration observed in the industrial transfer.

To construct the phase-level actual durations, the recorded duration of the parent transfer is distributed across its phases. First, the nominal durations of all phases belonging to the transfer are summed. A scaling factor is then calculated as:

$$\alpha = \frac{T_{\text{actual,transfer}}}{\sum_j T_{\text{nominal},j}}.$$

Each phase nominal duration is then multiplied by this factor:

$$T_{\text{actual},j} = \alpha T_{\text{nominal},j}.$$

In this way, phases with a larger nominal duration receive a larger share of the observed transfer time. The generated phase durations therefore remain consistent with the duration of

their parent transfer.

A small random variation of $\pm 10\%$ is then applied to the generated phase durations. This avoids producing identical execution patterns for phases with the same reference values and introduces controlled variability into the synthetic dataset.

The generator also introduces controlled error and retry behaviour. A failed phase can be followed by another attempt of the same phase. Failed attempts use a duration between 1.0 and 2.0 times their nominal duration.

Phase-attributed downtime Existing downtime events are preserved. They are associated with a mission phase only when the available context supports a reliable attribution. Events that cannot be assigned to one phase remain classified as between-phase downtime.

The main phase associations are summarised in Table 3.14.

Table 3.14: Main downtime sources attributable to mission phases

Phase	Main attributable downtime sources
MP1	Cross-walk and obstacle-related interruptions
MP2	Manoeuvre-area, path, obstacle, swap body, and spot-related interruptions
MP3	Gate, swap body, and spot-related interruptions
MP4	Gate and swap body handling interruptions
MP5	Gate, obstacle, and spot-related interruptions
MP6	Swap body handling interruptions
MP7	Manoeuvre-area, obstacle, and visibility-related interruptions
MP8	Path, obstacle, and visibility-related interruptions

When downtime is attributed to a phase, its duration contributes to the phase Availability calculation. Between-phase downtime is kept outside the phase metric and remains available at transfer level.

Phase-specific speed losses Temporary execution delays are represented separately through **Speed_Loss_Events**. These events do not reduce phase Availability. Instead, they increase the actual execution duration and therefore affect ESR.

Speed-loss types depend on the operational function of the phase. MP4 and MP6 are stationary handling phases and do not receive generated speed-loss events. The remaining phases can receive movement- or manoeuvring-related losses. It is summarised in Table 3.15.

Table 3.15: Speed-loss types by mission phase

Phase	Description	Speed-loss types
MP1	Follow the predefined path	Intersection wait; protective stop; moving-obstacle wait
MP2	Manoeuvre toward pre-insertion point	Protective stop; remote validation; intersection wait; moving-obstacle wait
MP3	Insert into spot under swap body	Remote validation; protective stop; slow-approach correction
MP4	Pick up swap body	None – stationary phase
MP5	Insert into empty spot	Remote validation; protective stop; slow-approach correction
MP6	Set down swap body	None – stationary phase
MP7	Extract from spot	Protective stop; remote validation; moving-obstacle wait
MP8	Catch up the predefined path	Intersection wait; protective stop; moving-obstacle wait

This separation preserves the distinction between Availability and Performance losses. Downtime reduces phase Availability, whereas temporary execution delays appear through ESR.

The resulting dataset contains 8,035 mission-phase records representing 1,561 missions across 800 expanded transfers. A total of 597 synthetic speed-loss events are generated. Transfers that cannot be expanded remain available only at transfer level.

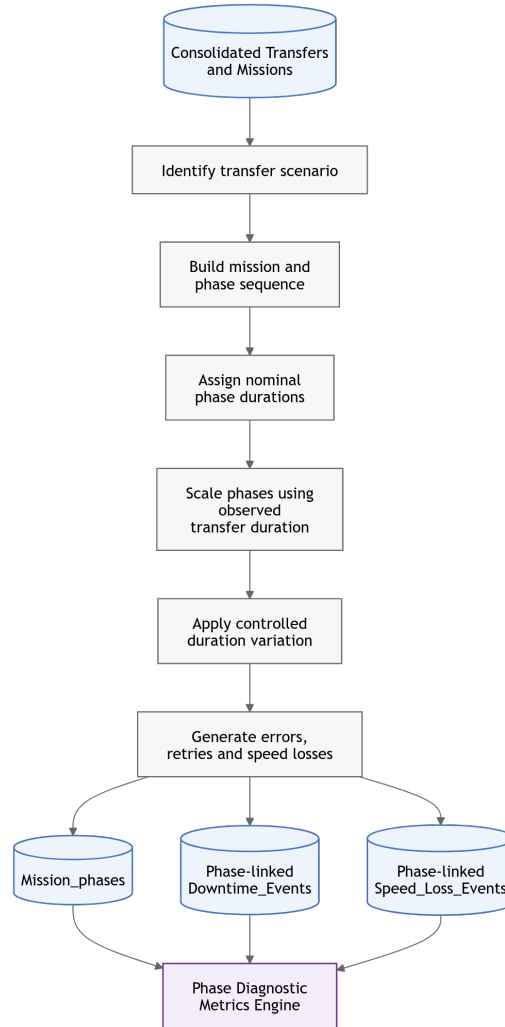


Figure 3.5: Construction of the synthetic mission-phase dataset

3.5.4 Phase-Level Diagnostic Metrics

The generated mission-phase dataset is used to calculate diagnostic Availability, Performance, and Quality for each phase.

Phase Availability considers only downtime that can be assigned to a specific phase:

$$A_{\text{phase}} = 1 - \frac{T_{\text{DT,phase}}}{T_{\text{phase}}}.$$

Downtime that cannot be linked to a phase remains classified as between-phase downtime. It is retained only in the transfer-level calculation.

At mission-phase level, Performance is represented by the Execution Speed Rate (ESR) only:

$$P_{\text{phase}} = \text{ESR}_{\text{phase}}.$$

DUR is not used at this level. DUR measures the share of vehicle Available Time occupied by qualifying execution. This denominator is meaningful at transfer and operating-period level, but not for an individual phase. For a short phase, a phase-level DUR would mainly reflect how often that phase occurs during the shift rather than how efficiently it is executed.

ESR is therefore more suitable for phase diagnosis. It directly compares the actual duration of a phase with its nominal reference and allows MP1–MP8 to be compared independently of their frequency or duration within the shift.

The phase-level Quality indicator follows the time-based Quality principle defined in Chapter 2. Availability, ESR, and Quality can then be combined into a diagnostic phase effectiveness value:

$$\text{OEE}_{\text{phase,diag}} = A_{\text{phase}} \times \text{ESR}_{\text{phase}} \times Q_{\text{phase}}.$$

This value is used only to compare phases. It is not aggregated to obtain the headline OEE. Table 3.16 presents the diagnostic metrics obtained for the generated mission-phase dataset.

Table 3.16: Diagnostic metrics by mission phase

Phase	Qualifying instances	Availability (%)	ESR (%)	Quality (%)	Diagnostic phase OEE (%)
MP1	859	85.49	43.16	96.19	35.40
MP2	1,392	36.52	14.22	93.39	4.85
MP3	594	42.68	3.61	89.56	1.38
MP4	537	63.22	2.57	80.07	1.30
MP5	667	44.15	12.38	91.60	5.01
MP6	527	86.35	5.02	87.67	3.80
MP7	1,560	96.57	6.26	97.69	5.91
MP8	908	89.24	26.39	97.36	22.93

The generated dataset shows different loss patterns across the mission sequence. MP2 has the lowest phase Availability, while MP3 and MP4 have the lowest ESR values. MP4 also has the lowest generated Quality value.

The mission-phase extension therefore focuses on Availability, ESR, Quality, and diagnostic phase effectiveness. These indicators demonstrate how the framework can localise losses within the operational sequence without modifying the authoritative transfer-level OEE.

3.5.5 Phase Execution-Time Capability

Phase-level capability analysis was applied to assess the dispersion of reconstructed execution times relative to their nominal references. The analysis follows the same methodology introduced in Section 2.10.6 and uses an analytical tolerance of $\pm 30\%$ around the nominal duration. Accordingly, execution-time ratios between 0.70 and 1.30 are considered within the selected tolerance.

Table 3.17 presents the capability results obtained for the eight mission phases.

Table 3.17: Mission-phase execution-time capability

Phase	n	Mean actual-to-nominal ratio	Standard deviation	Mean duration (s)	C_p	C_{pk}
MP1	825	2.47	13.18	154.1	0.0076	-0.0297
MP2	1,300	6.33	27.15	284.7	0.0037	-0.0617
MP3	532	24.95	109.66	249.5	0.0009	-0.0719
MP4	430	28.45	119.44	142.2	0.0008	-0.0758
MP5	611	8.66	67.23	86.6	0.0015	-0.0365
MP6	462	9.61	71.42	28.8	0.0014	-0.0388
MP7	1,524	14.76	84.30	147.6	0.0012	-0.0532
MP8	884	3.79	27.29	75.8	0.0037	-0.0304

All phases exhibit very low C_p values and negative C_{pk} values. The mean execution-time ratio therefore lies above the selected upper tolerance for every phase. The largest deviations from the nominal reference are observed for MP3 and MP4, whereas MP1 and MP8 remain comparatively closer to their reference durations.

These results are based on reconstructed phase-level durations. They are therefore used to demonstrate the capability-analysis function of the diagnostic layer rather than to characterise the measured capability of the deployed system. A direct industrial assessment would require phase-resolved timestamps recorded during operation.

3.6 Validation Approach

The industrial application presented in the previous sections shows that the proposed framework can be applied to heterogeneous operational records. However, the industrial dataset does not

provide a reference OEE against which the calculated result can be directly verified. In addition, several inputs required by the framework are reconstructed, statistically derived, or configured because they were not continuously recorded during operation.

Controlled synthetic data are therefore used to complement the industrial application. At transfer level, datasets with predefined operating conditions are generated and processed through the same metrics engine. This makes it possible to compare the calculated Availability, Performance, Quality, and OEE with expected values. At mission-phase level, the reconstructed MP1–MP8 structure is used to examine whether phase-specific losses are reflected in the corresponding diagnostic indicators.

The three parts of the validation approach are summarised in Figure 3.6.

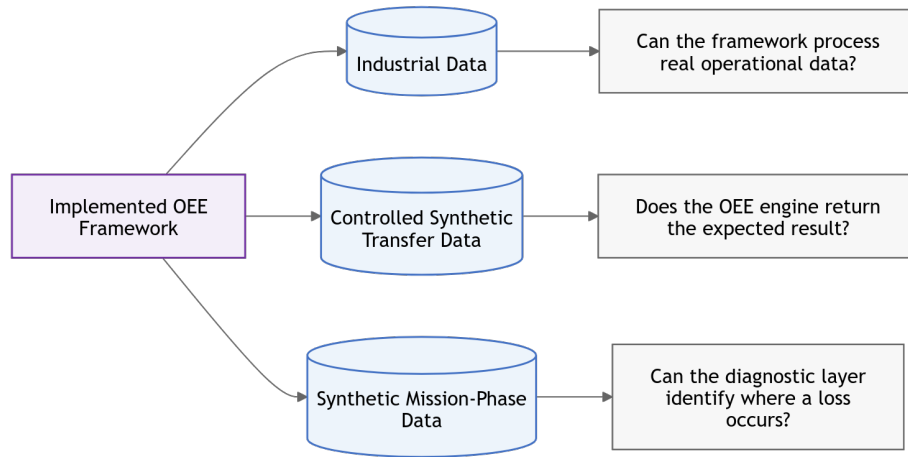


Figure 3.6: Validation approach of the implemented OEE framework

The industrial-data branch evaluates practical applicability, while the controlled datasets provide numerical and diagnostic validation of the transfer- and phase-level calculations.

3.6.1 Validation Checks

For every controlled dataset, the calculated OEE must satisfy the relationship

$$\text{OEE} = \frac{T_u}{T_r} = A \times P \times Q.$$

The Performance calculation must also preserve the decomposition

$$P = \text{DUR} \times \text{ESR}.$$

The difference between the equivalent calculation paths is required to remain below 10^{-6} . The calculated time levels must also preserve the six-level hierarchy

$$T_t \geq T_o \geq T_r \geq T_a \geq T_n \geq T_u \geq 0,$$

and the three OEE factors must remain within

$$A, P, Q \in [0, 1].$$

For scenarios generated around predefined numerical targets, a maximum deviation of five percentage points is accepted:

$$|X_{\text{computed}} - X_{\text{target}}| \leq 5 \text{ percentage points},$$

where X represents A , P , Q , or OEE.

The main checks applied during validation are summarised in Table 3.18.

Table 3.18: Validation checks applied to the implemented framework

Validation check	Expected result
OEE calculation	$T_u/T_r = A \times P \times Q$, deviation $< 10^{-6}$
Performance calculation	$P = \text{DUR} \times \text{ESR}$, deviation $< 10^{-6}$
Time hierarchy	Each time level remains non-negative and below its parent
Factor range	$A, P, Q \in [0, 1]$
Controlled numerical target	Deviation ≤ 5 percentage points
Factor sensitivity	Increasing a loss decreases its corresponding factor
Mission-phase diagnosis	A phase-specific loss is reflected in the corresponding phase indicator

3.7 Controlled Synthetic Validation

The industrial application presented in the previous sections demonstrates that the framework can process heterogeneous operational records, but it does not provide a known reference OEE. Controlled synthetic datasets were therefore used to evaluate the implemented metrics engine under predefined operating conditions. Three validation profiles were considered—Degraded, Operational, and High-OEE—and processed through the same calculation pipeline, with Availability, Performance, and Quality controlled through their corresponding loss mechanisms.

3.7.1 Synthetic Data Generator and Validation Profiles

A configurable synthetic-data generator was developed to create the datasets required for validation. The generator produces operating schedules, transfer records, downtime events, execution outcomes, and the additional inputs required by the OEE calculation.

Its interface allows the main simulation parameters and operating conditions to be configured, including the simulation horizon, fleet size, transfer activity, nominal duration, target Availability, Performance and Quality, weather conditions, speed losses, excluded time, and downtime distribution. A random seed can also be specified, and configurations can be stored as reusable validation profiles.

Figure 3.7 shows the implemented interface using one of the validation profiles.

OEE Mock Data Generator

Configure a profile → Save → Generate. Results go to `mock_ee_data/results/`.

Profile name: Description:

Simulation Parameters

Seed: Start date (YYYY-MM-DD): Num days: Fleet size:

Transfers/vehicle/day: Nominal duration (sec):

OEE Targets

Availability (A): Performance (P): Quality (Q):

OEE (A × P × Q): **76.97%**

Weather Mode

Weather mode:

Real weather data is used. ODD violations reduce Tr each day — both the generator and engine apply the same thresholds, so targets are still hit exactly.

Speed Loss

Speed loss fraction:

Excluded Time (avg sec / vehicle / day)

These subtract from Tr before Availability is computed. Each value is the average across the fleet per day — the generator samples randomly so some days have more, some have less (exponential distribution).

SW/FW update (sec): Charging (sec): Planned maint. (sec): E-stop human (sec):

Downtime split (active when A < 1)

System Fault: Operational Error: Intervention:

Figure 3.7: Configuration interface of the synthetic-data generator

The three OEE dimensions are controlled through their corresponding loss mechanisms. Availability is influenced through unplanned downtime, Performance through the relationship between qualifying nominal workload and Available Time, and Quality through the share of qualifying execution contributing to Useful Time.

The Required Time is calculated from the operating schedule generated for each profile and is therefore not assumed to be identical across the validation datasets. The target values used to configure the three profiles are summarised in Table 3.19.

Table 3.19: Target values of the validation profiles

Validation profile	Target A	Target P	Target Q	Target OEE
Degraded	$\approx 80\%$	$\approx 74\%$	$\approx 85\%$	$\approx 50\%$
Operational	$\approx 93\%$	$\approx 91\%$	$\approx 95\%$	$\approx 80\%$
High-OEE	$\approx 100\%$	$\approx 100\%$	100%	$\approx 100\%$

The target values of the individual OEE factors are used to define the intended operating condition. The numerical acceptance criterion introduced in Section 3.6.1 is applied to the headline OEE value.

3.7.2 Cross-Profile Comparison

The three validation profiles provide a controlled progression from a loss-intensive operating condition to one approaching the upper effectiveness range. The principal results are summarised in Table 3.20.

Table 3.20: Cross-profile comparison of the controlled validation results

Profile	Target OEE	A	P	Q	Computed OEE	OEE deviation
Degraded	50.00%	74.62%	78.66%	85.67%	50.29%	+0.29 pp
Operational	80.00%	94.50%	91.84%	92.10%	79.93%	−0.07 pp
High-OEE	100.00%	99.00%	99.49%	100.00%	98.49%	−1.51 pp

The Degraded Profile reaches an OEE of 50.29%. Losses are present across all three factors, with Availability showing the strongest reduction. The Operational Profile reaches 79.93%, as the losses in Availability, Performance, and Quality are substantially reduced. The High-OEE Profile reaches 98.49%. Quality reaches 100%, while small residual losses remain in Availability and Performance.

All computed OEE values remain within the five-percentage-point acceptance tolerance defined in Section 3.6.1. The progression of the individual factors and the resulting OEE is illustrated in Figure 3.8.

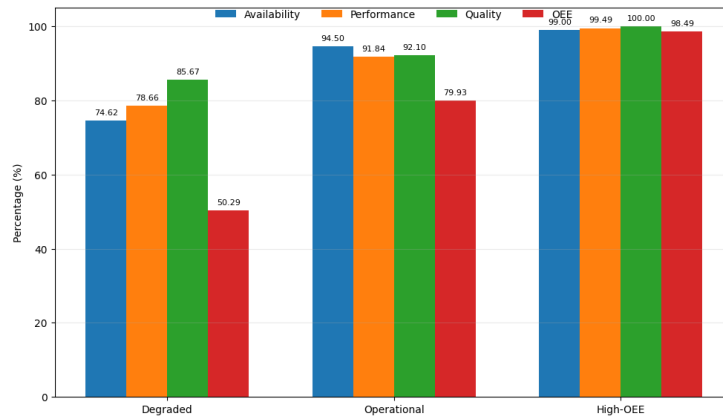


Figure 3.8: Comparison of Availability, Performance, Quality, and OEE across the profiles

Execution-time capability was additionally evaluated for each transport scenario across the three validation profiles. This analysis complements the OEE comparison by examining the dispersion and centring of execution times relative to their nominal references. The resulting C_{pk} values are presented in Table 3.21.

All transport scenarios exhibit positive C_{pk} values well above unity across the three profiles, indicating that the generated execution-time distributions remain centred within the selected

Table 3.21: Execution-time capability across the validation profiles

Transport scenario	Degraded	Operational	High-OEE
Close-proximity gate	17.282	18.096	43.366
Close-proximity parking spot	8.579	9.415	23.431
Park	5.698	3.246	9.658
RMF parking spot	9.314	9.359	22.465

tolerance around their nominal references. The High-OEE Profile generally produces the highest capability values, reflecting the lower execution-time variability configured for this operating condition.

The capability results do not, however, follow the OEE level directly. OEE combines losses in Availability, Performance, and Quality, whereas C_{pk} characterises the distribution of execution times for a given transport scenario. A profile may therefore exhibit reduced overall effectiveness because of downtime, unsuccessful transfers, or other losses while maintaining consistent execution times for successfully completed operations. This explains why the Degraded Profile can retain high capability values despite its lower OEE.

The comparison therefore confirms that the framework distinguishes global operational effectiveness from execution-time consistency. The controlled profiles produce the expected variation in OEE while capability remains an independent diagnostic measure of execution-time behaviour.

3.7.3 Downtime Diagnosis Across the Validation Profiles

The downtime reporting layer was also examined across the three controlled profiles. Table 3.22 compares the generated downtime durations and the effective Availability losses used by the metrics engine.

Table 3.22: Cross-profile comparison of generated and effective downtime

Profile	System fault (h)	Operational error (h)	Intervention (h)	Generated downtime (h)	Effective downtime (h)
Degraded	94.080	39.200	23.117	156.397	113.313
Operational	15.680	9.408	6.272	31.360	24.572
High-OEE	3.938	2.362	1.575	7.875	7.875

The generated downtime decreases substantially from the Degraded to the High-OEE Profile. For the Degraded and Operational profiles, the effective Availability loss is lower than the raw generated downtime because downtime is limited to the Required Time available on each operating day.

The comparison also highlights the distinction between event frequency and downtime severity. Although the High-OEE Profile contains a relatively large number of short downtime events, their accumulated duration remains small and therefore has only a limited effect on Availability.

3.7.4 Factor Sensitivity Check

The three validation profiles evaluate the complete data-generation and computation pipeline at selected operating levels. A one-parameter-at-a-time sensitivity analysis was additionally performed to examine how the individual OEE factors respond to their corresponding losses.

In each test, one loss mechanism was varied from 0% to 30%, while the remaining conditions were kept constant. Unplanned downtime was varied for Availability, unused Available Time for Performance, and unsuccessful execution for Quality.

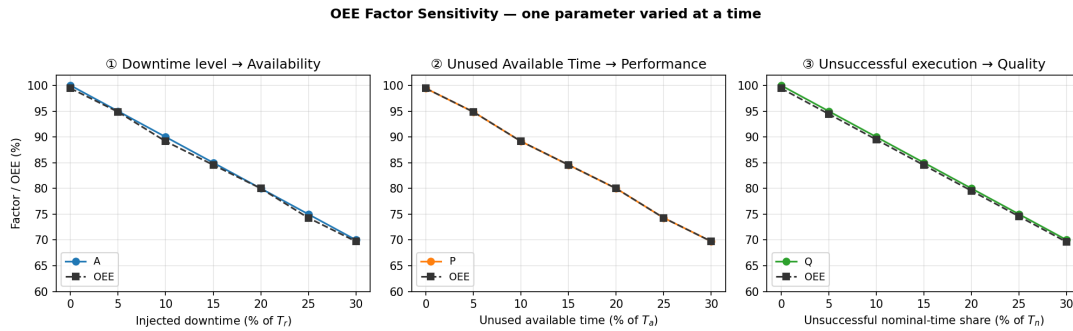


Figure 3.9: One-parameter-at-a-time sensitivity of the OEE factors

Increasing each imposed loss produces the expected decrease in its corresponding factor. Availability decreases as unplanned downtime increases, Performance decreases as the unused share of Available Time increases, and Quality decreases as unsuccessful execution increases. The remaining factors stay unchanged within each test.

In the Availability and Quality tests, the OEE curve lies slightly below the varied factor because the reference Performance remains marginally below 100%. Transfers are scheduled as complete operational units, and the remaining Available Time at the end of an operating window may be insufficient to accommodate another complete transfer. This residual unused time slightly reduces DUR while ESR remains at its reference level.

In the Performance test, Availability and Quality remain at 100%, and OEE therefore coincides with Performance.

The sensitivity analysis confirms the expected response of each OEE dimension to its associated loss. Together with the controlled validation profiles, the results show that the implemented framework behaves consistently across different operating levels and across a continuous range of controlled loss conditions.

3.8 Mission-Phase Diagnostic Validation

The controlled validation in Section 3.7 verified the transfer-level OEE engine against predefined Availability, Performance, and Quality conditions. The mission-phase layer requires a different validation approach. Its phase durations are reconstructed from the parent transfer duration and therefore do not have an independent measured phase-level ground truth.

The three controlled datasets introduced in Section 3.7 were processed through the mission-phase pipeline. Each dataset was expanded into MP1–MP8 records using the reconstruction

procedure defined in Section 3.5.3. The resulting records were then evaluated using the phase diagnostic engine introduced in Section 3.5.4. Figure 3.10 summarises the validation procedure.

Mission-Phase Diagnostic Validation Flow

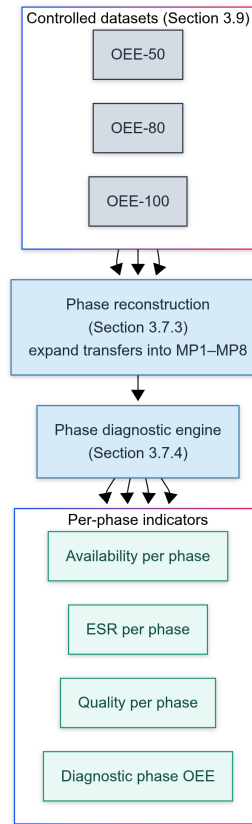


Figure 3.10: Mission-phase diagnostic validation flow

3.8.1 Controlled Phase-Level Datasets

The three controlled datasets were expanded using the same phase-reconstruction procedure. Each generated phase record contains its nominal reference, reconstructed duration, outcome, and any downtime that can be associated with that phase.

The resulting datasets contain 16,415, 22,346, and 25,617 mission-phase records for the OEE-50, OEE-80, and OEE-100 profiles, respectively. These dataset sizes provide the basis for the phase-level comparisons presented below.

The three datasets therefore provide a controlled progression from a degraded condition to a nearly ideal operating condition.

3.8.2 Phase Availability Response

Phase Availability provides a clear diagnostic response to the controlled degradation. At the OEE-100 level, all phases reach full Availability because no phase-attributed downtime is generated.

As the overall effectiveness decreases, the largest Availability losses appear in MP2, MP7, and MP8, with an additional reduction in MP5 at the OEE-50 level. MP2 and MP7 show the

strongest degradation, reaching 37.70% and 49.48%, respectively. MP8 decreases to 73.58

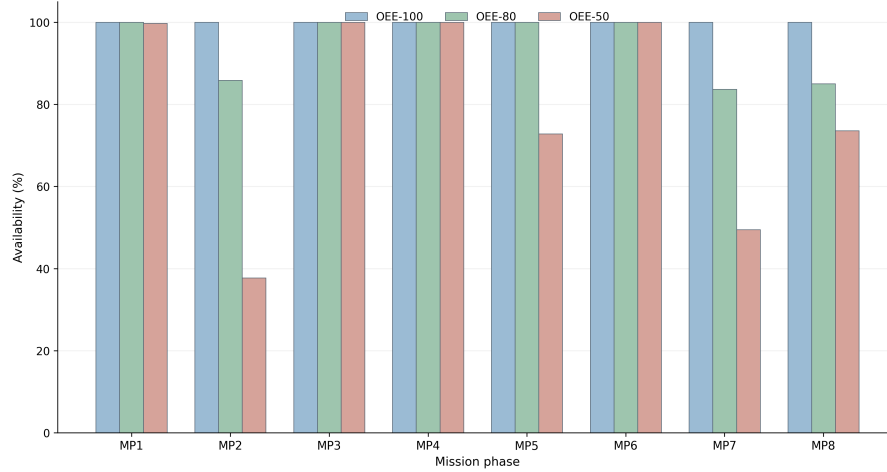


Figure 3.11: Phase Availability across the controlled validation levels

The observed Availability pattern is consistent with the role of the different mission phases. MP2, MP7, and MP8 involve manoeuvring, extraction, and return-to-path movements. These phases are more exposed to interruptions during vehicle motion and positioning. MP5 is also affected at the OEE-50 level because it involves insertion into an empty spot.

In contrast, MP4 and MP6 are short stationary handling actions. Their Availability remains at 100% because no downtime is associated with these phases in the controlled datasets.

The pattern is therefore operationally plausible. However, it should not be interpreted as proof that these phases are the least available in real operation. The main result is that the diagnostic localises the introduced losses and shows their increasing severity as the overall effectiveness decreases.

3.8.3 Phase Quality Response

Phase Quality also responds consistently to the controlled degradation, but the loss is localised to different parts of the mission sequence. Figure 3.12 presents the progression.

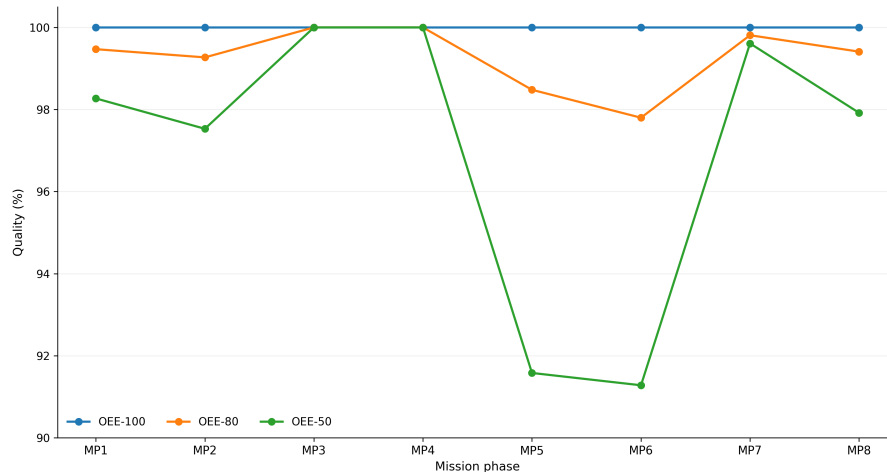


Figure 3.12: Phase-level Quality across the controlled validation profiles

At the OEE-100 level, all phases reach 100% Quality because no unsuccessful phase execution is introduced. As the effectiveness level decreases, the strongest Quality losses appear in MP5 and MP6. At the OEE-50 level, their Quality values fall to 91.58% and 91.28%, respectively. MP1, MP2, MP7, and MP8 show smaller reductions. MP3 and MP4 remain at 100% across all three levels.

The strongest Quality losses occur in MP5 and MP6, corresponding to the final insertion and set-down part of the transfer. Failures occurring at this stage affect the final transfer outcome while earlier phases, including MP3 and MP4, remain classified as successfully completed.

Phase-level Quality therefore remains higher than transfer-level Quality because unsuccessful execution is localised to the phase where the failure occurs rather than propagated to the complete mission sequence. Previously completed phases remain successful. Phase Quality should consequently be interpreted as a localisation indicator that identifies where unsuccessful execution becomes visible within the mission sequence, rather than as a second measure of complete-transfer success.

3.8.4 Phase Performance Response

At mission-phase level, Performance is represented by the Execution Speed Rate (ESR). Table 3.23 presents the values obtained across the three controlled validation levels.

Table 3.23: Phase Execution Speed Rate across the validation profiles

Phase	OEE-100 (%)	OEE-80 (%)	OEE-50 (%)
MP1	95.16	94.99	94.61
MP2	95.22	94.88	94.08
MP3	95.14	95.28	95.25
MP4	95.12	95.23	95.28
MP5	95.22	94.53	91.63
MP6	95.30	94.44	92.09
MP7	95.23	95.07	95.05
MP8	95.19	94.98	94.25

The phase ESR varies less strongly than Availability and Quality. Most values remain between approximately 91% and 95%. The largest reductions at the OEE-50 level occur in MP5 and MP6.

MP5 shows the largest reduction among the moving phases. It corresponds to insertion into an empty spot and can receive several manoeuvring-related speed losses. Its ESR decreases from 95.22% at OEE-100 to 91.63% at OEE-50.

MP6 follows a different pattern. It is a stationary set-down phase and does not receive generated speed-loss events. Its execution time can nevertheless increase when a set-down attempt fails or must be repeated. Its ESR therefore decreases from 95.30% to 92.09%.

MP4 is also stationary and is excluded from the speed-loss mechanism. Its ESR consequently

remains almost unchanged across the three levels. The remaining phases show only small changes because the additional execution time is limited when averaged across all qualifying phase executions.

3.8.5 Diagnostic Phase Effectiveness

The diagnostic phase effectiveness combines phase Availability, ESR, and Quality into a single indicator. It therefore shows where several loss mechanisms accumulate within the mission sequence. Figure 3.13 presents the results for the three validation profiles.

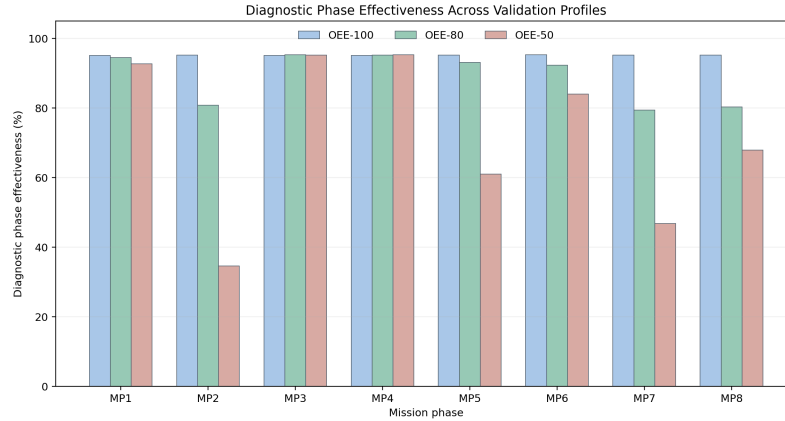


Figure 3.13: Diagnostic phase effectiveness across the controlled validation profiles

At the OEE-100 level, all phases remain close to 95%. Availability and Quality are equal to 100

As the overall effectiveness decreases, the phase pattern becomes more differentiated. MP2 and MP7 show the strongest reductions. At the OEE-50 level, their diagnostic effectiveness falls to 34.59% and 46.85%, respectively. MP5 and MP8 are also strongly affected.

The low value of MP2 is mainly driven by its Availability loss. The same applies to MP7. MP5 combines a lower Availability with reductions in ESR and Quality. MP6 keeps full Availability, but its Quality and ESR decrease. The combined indicator therefore shows not only where a loss occurs, but also where several loss dimensions affect the same phase.

MP3 and MP4 remain close to their OEE-100 values across the three levels. These phases receive little or no Availability and Quality loss in the controlled datasets. Their diagnostic effectiveness is therefore determined mainly by ESR.

The phase-level view therefore adds information that is not available from the transfer-level OEE alone. The transfer-level indicator quantifies the overall effectiveness loss. The phase diagnostic identifies where the losses occur within the mission sequence.

The diagnostic phase effectiveness is used only for phase comparison and loss localisation. It is not aggregated to reproduce the transfer-level OEE.

3.9 Discussion and Limitations

3.9.1 Overall Findings and Root-Cause Synthesis

The implementation demonstrates that the proposed OEE framework can be applied to autonomous yard-logistics operations using heterogeneous industrial records. The automatic data export and manual field logs were harmonised, reconstructed, and mapped to the canonical data model before the OEE indicators were computed. The canonical model therefore provides a common structure between the available operational data and the metrics engine.

The industrial application also confirms the need to separate different types of operational loss. Availability accounts for unplanned interruptions during Required Time, Performance reflects additional execution time relative to the nominal reference, and Quality distinguishes useful autonomous completion from unsuccessful or manually completed execution. The complementary reliability, process-quality, capability, and supervision indicators provide additional diagnostic information without modifying the OEE calculation itself.

The industrial results should nevertheless be interpreted as an illustrative application of the framework rather than as a definitive measurement of deployed vehicle performance. Several required inputs were not recorded directly and had to be reconstructed, derived, or configured. The industrial analysis therefore demonstrates the complete calculation chain and highlights the information required for a more reliable future deployment.

The industrial database provided by the company consisted of test data intended to validate the applicability of the proposed methodology, not to evaluate actual operational performance. The dataset was inherently limited: it covered a single vehicle over a restricted recording period, which constrains how representative the results can be of broader fleet or long-term behaviour. The calculated values should therefore be treated as diagnostic baseline indicators rather than as evidence of achieved improvement. Given these limitations, the low measured effectiveness — Availability of **57.08%**, Performance of **21.62%**, Quality of **59.08%**, and an **OEE of 7.29%** across **908 transfers** and **1,692 missions** — is not unexpected. It reflects the constrained scope and quality of the available recordings as much as it reflects the vehicle’s actual behaviour. These values are nonetheless useful because they show how the framework quantifies the current loss structure and where downtime and speed losses concentrate: system faults accounted for **69.73%** of downtime, recovery gaps for a further **24.02%**, and the low Performance value was mainly associated with an Execution Speed Rate of **21.62%**. These findings indicate where improvement actions should be investigated. A definitive operational interpretation would require data collected across more vehicles and over a longer, more consistently recorded period.

In contrast, the controlled synthetic validation demonstrates that the calculation engine itself behaves correctly under known conditions. The degraded, operational, and high-effectiveness profiles produced OEE values close to their intended targets: **50.29%**, **79.93%**, and **98.49%**, respectively. The sensitivity analysis confirmed that controlled changes in Availability, Performance, and Quality propagate correctly to the corresponding factors. This shows that the

framework works as designed for a given scenario and correctly captures and reproduces effectiveness losses when the input data are complete and reliable. These synthetic results validate the engine’s internal consistency. They are not measurements of real-world improvement and must not be conflated with the industrial findings above.

The mission-phase extension adds a second level of analysis. Transfer-level OEE provides the overall effectiveness of the transport operation, while phase-level Availability, ESR, Quality, and diagnostic phase effectiveness localise losses within the MP1–MP8 sequence. The two levels are therefore complementary: the transfer level indicates how much effectiveness is lost, while the phase level helps identify where the loss occurs.

The different factors identified throughout the analysis are summarised in the Ishikawa diagram shown in Figure 3.14. The diagram groups the potential causes of reduced operational effectiveness into six categories: vehicle and autonomy behaviour, mission execution, yard and traffic interaction, human supervision, operational coordination, and environmental conditions.

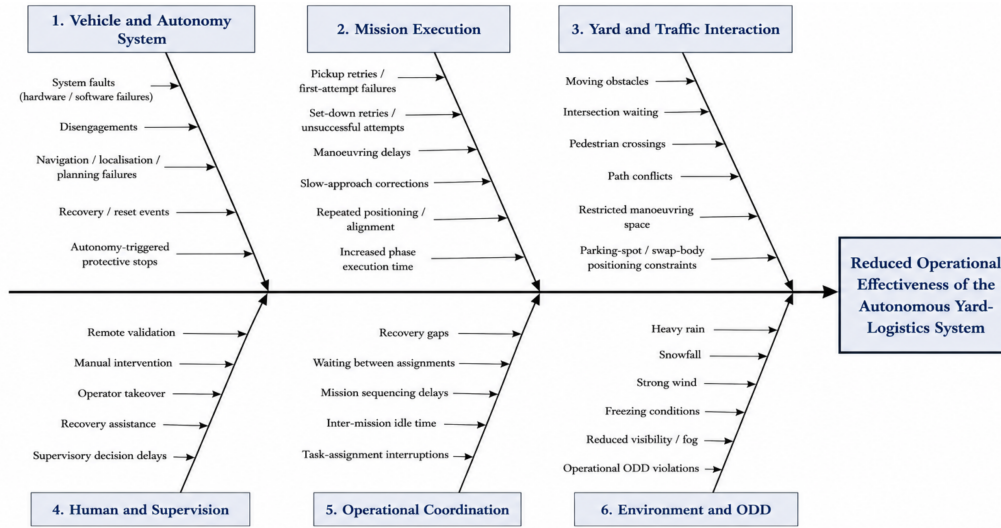


Figure 3.14: Root-cause synthesis of factors affecting operational effectiveness in autonomous yard logistics.

The synthesis shows that reduced effectiveness cannot be attributed to a single technical cause. System faults and recovery-related events mainly contribute to Availability losses. Protective stops, manoeuvring delays, and other increases in execution time affect Performance. Retries and unsuccessful mission execution contribute to Quality losses. Human supervision, yard interactions, operational coordination, and ODD restrictions may influence one or more dimensions. The proposed framework combines these views by quantifying their global effect at transfer level and supporting a finer diagnosis at mission-phase level.

3.9.2 Limitations and Future Data Requirements

The main limitations can be divided into three categories: data completeness, telemetry and measurement, and loss classification.

3.9.2.1 Data Completeness

The main limitation of the industrial evaluation is the **quality and completeness of the available data**. The automatic data export and manual field logs were produced independently and differ in structure, identifiers, temporal precision, and acquisition periods. As a result, only part of the observations could be matched directly, while the remaining records required source-specific reconstruction.

This limitation is consistent with the purpose of the industrial database: it was made available to test the methodology, not to evaluate operational performance. Consequently, the reported indicators demonstrate that the calculation chain can be executed on real operational records, but they cannot by themselves establish a statistically reliable improvement in availability, performance, or quality.

Several inputs required by the OEE framework were also unavailable as direct measurements. The operating schedule and planned exclusions were configured from project information, while nominal transfer durations were derived from successful executions. Some repair and event durations were also missing and therefore required fallback values. These assumptions make the calculation possible, but they also introduce uncertainty into the results.

The environmental analysis has a similar limitation. Weather conditions were reconstructed from ERA5 hourly data using the closest available grid point rather than measurements collected directly at the operating yard. The resulting weather-based ODD exclusions should therefore be considered an approximation of the actual environmental conditions experienced by the vehicle.

The mission-phase diagnostic layer is further limited by the absence of reliable timestamps for MP1–MP8 in the industrial records. Phase durations were therefore reconstructed using transfer and mission information together with configured nominal references. The phase indicators demonstrate the diagnostic potential of the framework, but they should not be interpreted as directly measured phase performance.

3.9.2.2 Telemetry and Measurement

The framework depends strongly on telemetry data. Telemetry provides the timestamps, states, durations, and event signals used to reconstruct the time hierarchy and the OEE factors. Data quality problems can therefore affect the results. These problems include missing messages, delayed transmission, unsynchronised clocks, coarse sampling, and changes in the meaning of status codes.

The framework measures what can be observed through the telemetry system. It cannot identify losses that are not recorded or cannot be inferred. This dependence limits comparisons between different deployments unless the data schema, timestamp rules, sampling resolution, and event definitions are standardised and carefully controlled.

3.9.2.3 Loss Classification

Loss classification is another important limitation. Some events cannot be assigned to a single category directly. For example, a single event may be related to technical downtime, an autonomy limitation, or an ODD exclusion. The same event may also affect more than one mission phase.

The proposed precedence and overlap-correction rules make the calculation more consistent, but engineering judgement is still required when defining event mappings. The classification system may also need to change as the system evolves. New failure codes, intervention modes, or operating procedures may require a revised taxonomy.

A stable classification dictionary is therefore needed. Clear event precedence rules should also be defined, and regular review with operations and maintenance experts is recommended.

3.9.2.4 Future Data Requirements

Future implementations should prioritise the direct and consistent recording of the inputs required by the proposed data model. The following information should be prioritised:

- persistent transfer and mission identifiers
- complete shift schedules and planned stoppages
- synchronised timestamps
- technical-failure and recovery timestamps
- structured ODD events
- start and end timestamps for each mission phase

Direct recording of these data would reduce the need for reconstruction and support continuous operational monitoring rather than only retrospective analysis.

3.9.3 Transferability to Other Environments

The framework is designed around concepts that can extend beyond the studied yard, including transfer-based measurement, hierarchical time accounting and ODD exclusions. It can therefore provide a useful starting point for other autonomous transport environments, such as different logistics yards, warehouses, or industrial sites. However, the current implementation should not be considered universally applicable without adaptation.

Portability requires recalibration of the mission and transfer definitions, nominal durations, outcome taxonomy and ODD thresholds. A warehouse may be dominated by congestion and pedestrian interactions, whereas a port may introduce different traffic rules, loading constraints, and weather limits. The meaning of a successful transfer, an acceptable intervention, or an excluded operating period may therefore differ between environments. In addition, a framework

validated with one vehicle, one telemetry architecture, and one site-specific dataset cannot by itself establish validity for another fleet or operating context.

Extension to other environments should therefore follow a staged process:

- define the local operational concepts and ODD criteria
- map available telemetry to the canonical data model
- calibrate nominal references and classification rules with domain experts
- validate the resulting indicators against independently observed operational outcomes

3.10 Conclusion

This chapter presented the implementation and validation of the proposed OEE framework for autonomous yard logistics. A structured ingestion pipeline was developed to transform heterogeneous industrial records into the canonical data model. Weather-based ODD information was then integrated before computing transfer-level OEE, complementary engineering indicators, and mission-phase diagnostic metrics.

Application to the consolidated industrial data demonstrated that the complete calculation chain can be executed using the proposed structure, while also revealing the effect of incomplete data on the interpretation of the results. **Controlled synthetic validation confirmed the consistency and sensitivity of the transfer-level calculations**, while the mission-phase analysis demonstrated the ability of the diagnostic extension to localise Availability, Performance, and Quality losses within the operational sequence.

Overall, the proposed framework combines a global measure of operational effectiveness with finer engineering diagnosis. More complete and directly measured operational data are required before the resulting indicators can be interpreted as definitive deployment-level performance measures. The main contribution is therefore a **structured and practical framework for measuring and diagnosing autonomous yard-logistics performance**.

General Conclusion

In this work, I studied how to evaluate the operational effectiveness of autonomous vehicles in yard-logistics environments. Classical OEE was designed for stationary manufacturing systems. It does not fully describe mission variability, autonomous driving constraints, remote supervision, or ODD conditions. I therefore proposed an OEE-based framework for autonomous transport operations.

I kept the classical Availability, Performance, and Quality structure. I selected the transfer as the main measurement unit. I also introduced a six-level time hierarchy. This hierarchy separates non-operating periods, planned and ODD exclusions, downtime, performance losses, and quality losses. I divided Performance into dispatch utilisation and ESR. I also added engineering indicators to support the diagnosis.

I developed a canonical data model and an ingestion pipeline for an incomplete industrial data. I applied the framework to an industrial dataset to demonstrate the calculation process. However, I had to reconstruct several inputs. The resulting indicators are therefore illustrative rather than definitive. I also used synthetic datasets to check the consistency of the calculations. Finally, I developed a mission-phase extension to identify losses at different stages of a mission.

The main limitations are related to data quality and event classification. Missing telemetry can affect the indicators. Ambiguous events can also lead to inconsistent classification. Clear classification rules and expert review are therefore necessary.

The main academic contribution of my work is the adaptation of OEE to autonomous yard logistics. The framework combines a global effectiveness indicator with engineering diagnostics. It also provides a common structure for heterogeneous operational data. Its industrial contribution is a practical method for identifying operational losses and supporting performance improvement.

Future work should improve data collection and reduce data reconstruction. Mission and transfer identifiers should be recorded directly. Timestamps, technical events, ODD conditions, and mission phases should also be synchronised and recorded consistently.

This work can provide a basis for standardising performance measurement for autonomous transport systems. However, I believe that further validation in different environments is required before defining a formal standard.

Overall, I developed a structured foundation for evaluating autonomous yard-logistics performance. The framework connects OEE, operational data, and engineering diagnostics in a common approach.

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REPUBLIC OF TUNISIA

Ministry of Higher Education and Scientific Research

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Design and Implementation of a Comprehensive OEE Framework for an Automated Transport System

Abstract

This graduation project presents an Overall Equipment Effectiveness (OEE) framework adapted to autonomous yard-logistics operations. The approach redefines Availability, Performance, and Quality for variable transport missions and integrates Operational Design Domain (ODD) constraints. A canonical data model and ingestion pipeline were developed to harmonise telemetry, manual logs, and weather data. The framework was implemented at transfer level and extended with mission-phase diagnostics. Industrial data demonstrated the calculation chain, while controlled synthetic scenarios validated the consistency and sensitivity of the implemented metrics.

Keywords: OEE, autonomous vehicle, yard logistics, ODD, canonical data model, mission-phase diagnostics

Conception et implémentation d'un cadre complet de TRS pour un système de transport automatisé

Résumé

Ce projet présente un cadre de Taux de Rendement Synthétique (TRS) adapté aux opérations de transport autonome en environnement logistique. L'approche adapte la Disponibilité, la Performance et la Qualité aux missions variables et intègre les contraintes ODD. Un modèle de données canonique et une chaîne d'intégration ont été développés pour harmoniser les données télémétriques, les journaux manuels et les données météorologiques. Le cadre est appliqué au niveau des transferts et complété par une analyse des phases de mission.

Mots-clés : TRS, véhicule autonome, logistique de parc, ODD, modèle de données canonique, phases de mission

تصميم وتنفيذ إطار شامل للفعالية الإجمالية للمعدات لنظام نقل آلي

ملخص

يقدم هذا المشروع إطاراً للفعالية الإجمالية للمعدات (OEE) مكيفاً مع عمليات النقل الذاتي في الساحات اللوجستية. يعتمد الإطار على تكييف مفاهيم الإتاحة والأداء والجودة مع طبيعة مهام النقل المتغيرة، مع دمج قيود مجال التشغيل التصميمي (ODD) كما تم تطوير نموذج بيانات موحد ومسار لمعالجة ودمج بيانات القياس الآلي والسجلات اليدوية والبيانات الجوية. طُبّق الإطار على مستوى عمليات النقل، مع إضافة طبقة تشخيصية على مستوى مراحل المهمة لتحديد مواقع الخسائر بشكل أدق. واستُخدمت البيانات الصناعية لإظهار سلسلة الحساب، بينما استُخدمت سيناريوهات اصطناعية مضبوطة للتحقق من اتساق المؤشرات وحساسيتها.

الكلمات المفتاحية: الفعالية الإجمالية للمعدات، المركبات الذاتية، لوجستيات الساحات، ODD نموذج البيانات، مراقبة الأداء، مراحل المهمة